

Review article

Advancing Network Security in Industrial IoT: A Deep Dive into AI-Enabled Intrusion Detection Systems

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ABSTRACT

The increasing use of Industrial Internet of Things (IIoT) devices has heightened concerns about cybersecurity threats, particularly botnet attacks. Traditional internet communication methods have consistently faced these challenges, leading to substantial economic losses for numerous manufacturing enterprises. As machine-to-machine communications grow, these attacks are becoming more prevalent. This research addresses the critical need for an AI-powered network intrusion detection system. We conducted an extensive literature review and implemented over 25 advanced Machine Learning (ML) algorithms, including various modifications, to detect botnet attacks on seven IoT devices. The primary objective was to develop robust and accurate models for identifying security threats and allow for a comprehensive performance benchmark for all these models utilizing the same dataset. Our findings revealed that certain models achieved near-perfect performance in detecting botnet attacks, while others were less effective. Our contributions include identifying high-performing ML models for botnet detection and demonstrating their applicability across various IoT devices. Future research should focus on validating these models with new datasets and exploring how the type and function of IoT devices influence detection performance and response time.

1. Introduction

As technology has rapidly progressed, the Internet has become an integral part of daily life, utilized for various personal and professional activities. This includes sensitive operations like communication, data exchange, and commercial transactions. While the Internet has been instrumental in bridging gaps and facilitating communication, its integrity is continually at risk from cyber-attackers who aim to compromise network security. The escalating frequency of these attacks underscores the urgency to develop more advanced protective measures. Every sector, from businesses to governments, requires fortified network security solutions, as no network is entirely impervious to cyber-attacks. In today's digital age, network security has evolved from a luxury to an urgent necessity for individuals, commercial entities, non-profit organizations, and government bodies. This transformation is driven by society's increasing reliance on many services operating on digital networks. Despite significant advancements in protective mechanisms, vulnerabilities that hackers exploit still exist, keeping network security professionals on their toes. Table 1 shows a summary of the parts of this paper.

Over recent years, there has hardly been a month without news of a major corporation in the retail or financial sector succumbing to cyber-attacks or data breaches that jeopardized customer information. Companies like Target, Home Depot, and Sony Pictures Entertainment were all subjected to cyber-attacks from 2013 to 2017 [1]. Stuxnet's 2010 assault on Iran's nuclear facilities remains one of the most infamous examples of how damaging cyber-physical attacks can be. The malware disrupted the operations of centrifuges by manipulating the data shown to the operators, effectively shortening the lifespan of the machinery [2,3]. In another historically significant case, the CIA orchestrated an attack in 1982 during the Cold War to disrupt the Soviet Union Trans-Siberian pipelines. Malicious code was surreptitiously placed into the control systems, causing unprecedented pressure levels and triggering a massive explosion observable from space [4,5,6,7]. The risks of cyber-physical attacks extend to healthcare as well. For example, the manipulation of insulin pump dosages can have fatal outcomes [8]. A German foundry experienced an attack on its blast furnace controls through targeted spear-phishing, resulting in severe damage and worker injuries [9,10,11]. The automotive and energy sectors have also been affected. The "WannaCry" ransomware disrupted production in multiple Renault

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Table 1
The structure of the paper.

Section	Subsection	Subdivision	Description
1	Introduction	NA	Cyber-attacks and why detecting threats is important
2	The Applications and Architecture of IoT	Perception Layer Network Layer Support Layer Application Layer	NA The different components of IoT devices and their potential
3	IoT in Manufacturing	NA	How IoT devices and sensors are being utilized in manufacturing
4	Taxonomy of IoT Attacks	The Distributed Denial of Service (DDoS) Phishing Attacks Malware Attack Botnet Attacks Zero-Day Attacks	NA Description of major attacks and their classification
5	Threat Detection Systems	NA	Discussion of different intrusion detection systems
6	The Available Datasets	NA	Description of the available cybersecurity datasets
7	Methodology	Background Utilized Algorithms	NA ML Algorithms DL Algorithms Dataset Preprocessing
8	Results and Discussions	Utilized Dataset Limitations Final Thoughts	NA Showing the results of the analysis and discussing its implications
9	Conclusion	NA	The final thoughts of the paper

factories across Europe [12,13,14]. A 2016 assault on Ukraine's power grid led to extensive outages [15]. These incidents underscore the vulnerability of industrial systems to cyber intrusions, with one-third of such attacks reportedly targeting the manufacturing sector [16,17]. Experiments by the Department of Energy's Idaho lab have also demonstrated the feasibility of attacks causing physical machine failures [18]. Moreover, cyber-attacks can have far-reaching impacts that don't immediately endanger lives but could have serious long-term consequences. For instance, altering collision airbag inspection results could lead to product recalls, significant financial loss, and reputational damage. Other potential attack vectors could include tampering with CAD files or altering manufacturing parameters [19]. Cyber-Physical Systems (CPS) is known as the system providing tight integration between real systems with cyberspace [20]. When CPS is connected to the internet, it is often referred to as the Internet of Things (IoT) [21]. The latter allows interaction and cooperation of inter-networked physical objects to collect and exchange data over the Internet [22]. Various methods have been proposed for classifying and mitigating network attacks. The port-based approach, which involves recognizing port numbers registered with the Internet Assigned Numbers Authority (IANA), has become less effective due to the growing multitude of applications that use dynamic or unregistered ports. Another method, Deep Packet Inspection (DPI), analyzes network packet contents against a known signature database [23]. Though more precise than the port-

based approach, DPI struggles with encrypted data and has a significant computational overhead. Behavioral classification techniques examine overall network traffic patterns to identify applications. While effective, these methods consume substantial resources and are susceptible to false positives. Techniques that focus on statistical characteristics of traffic flow also exist, but these too have limitations in terms of accuracy and computational efficiency [24]. Enter the age of Artificial Intelligence (AI) which showed immense promise in overcoming the drawbacks of traditional statistical classification methods. In the realm of network security, AI in the form of Machine Learning (ML) and Deep Learning (DL) algorithms can be trained to discern between benign and malicious network traffic. Moreover, they can identify specific types of cyber-attacks and trigger relevant defensive measures. The application of these intelligent techniques extends from large enterprises to law enforcement agencies and personal networks, offering a new layer of security. Historically, network attack detection relied heavily on pre-defined, signature-based databases that required frequent updates due to evolving attack vectors. The advent of newer ML and DL models has notably enhanced the predictive accuracy of identifying cyber threats, thus solidifying intelligent-based techniques as a burgeoning area of research in network security.

This paper tried to address the critical need for an AI-powered network intrusion detection system by conducting an up to date literature review followed by the implementation of over 25 advanced ML and DL algorithms, including various adjustments and hybrid modifications, to detect botnet attacks on seven IoT devices. The primary objective was to develop robust and accurate models for identifying security threats and allow for a comprehensive performance benchmark for all these models utilizing the same dataset to address a gap that was found in our literature review.

2. The applications and architecture of IoT

The IoT signifies a rapidly expanding domain focused on the autonomous exchange of information. This interconnected ecosystem encompasses objects equipped with a range of technologies such as sensors, control mechanisms, and software. Advancements in fields like ML and DL have catalyzed the development of IoT technology [25]. Its applications are now permeating diverse sectors. For instance, Fig. 1 illustrates numerous ubiquitous uses of IoT. Intelligent methods, such as ML and DL, have shown promise in various sectors like healthcare, finance, and education, among others. This has encouraged experts in network security to explore their potential further.

In today's digitally accelerated world, various industries are adopting connected technologies. The educational landscape, for example, has transcended traditional approaches; modern classrooms are digitally interconnected, facilitating learning for hearing-impaired students through connected devices like gloves and tablets. Likewise, IoT has substantial potential to aid students with other disabilities. On a broader scale, smart cities and homes are evolving to meet essential human needs including security, waste disposal, and air quality management, among other things [26]. Healthcare has undergone a significant transformation through IoT, with innovations ranging from wearable technology to remote patient monitoring [27]. In agriculture, IoT technologies have revolutionized traditional practices by enhancing water and soil management systems [28]. The automotive industry has witnessed groundbreaking changes through IoT-enabled connected vehicles [29]. and the energy sector has achieved new milestones in management with IoT integration in electrical grids [30]. A historical overview of IoT's growth is presented in Fig. 2. The technology has progressed from internet-capable refrigerators to comprehensive smart cities [31]. Milestones include its application in military strategy with the introduction of the Internet of Battlefields in 2017, and its integration into healthcare in subsequent years. Notably, the first Wi-Fi-enabled rabbit device was developed in Japan in 2005 [32].

The prominence of IoT in the Hype Cycle of emerging technologies

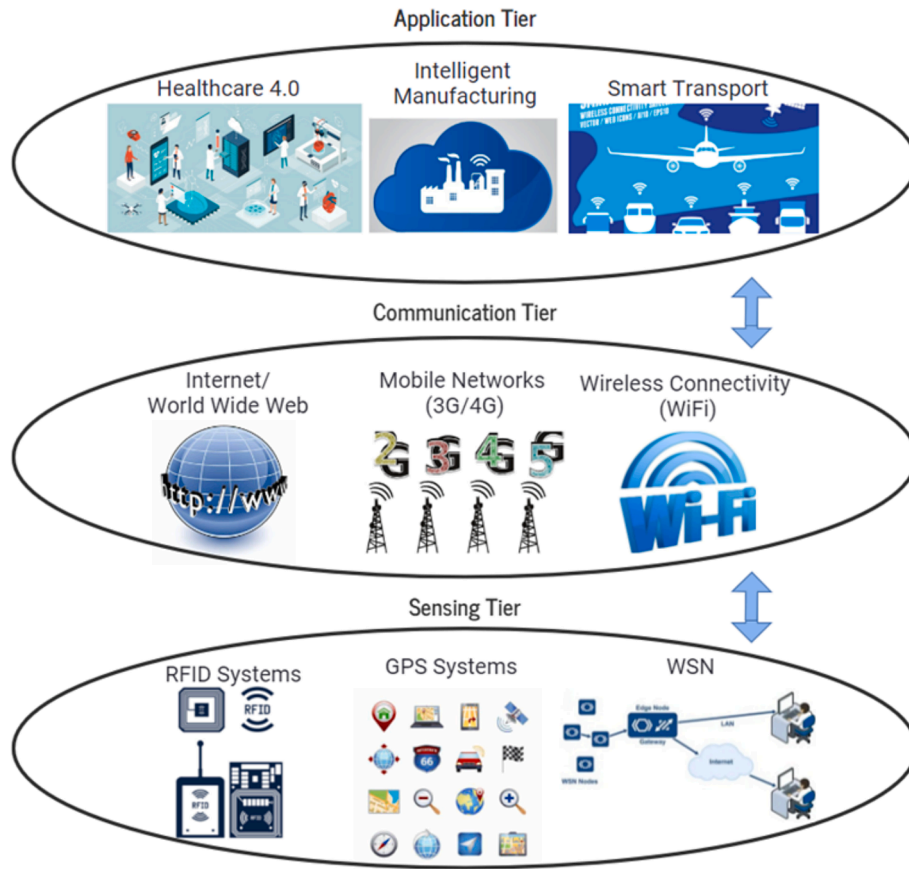


Fig. 1. Applications of IoT.

began in 2011, and it continues to dominate the cycle [33]. After the occurrence of a large-scale IoT-based attack in 2016, the emphasis has increasingly shifted toward enhancing IoT security [34]. Security measures are fragmented, with individualized solutions based on the device's manufacturer. Traditional security paradigms often prove ineffective for IoT devices, which typically feature limited processing capabilities. Other concerns include energy constraints and location-specific issues for implementing robust security measures. Mobility adds another layer of complexity, especially for IoT-enabled vehicles [35]. In addition, IoT presents various resource constraints, including issues related to cost, power, and size. The heterogeneity of devices, given their distributed nature involving sensors and actuators, further complicates matters [36]. Without standardized regulations, governments essentially leave IoT implementations to the discretion of service providers, complicating the establishment of a universal platform for IoT solutions [37]. IoT's efficacy is continually expected to improve with advancements in AI and data engineering. Interoperability remains a crucial need, given the rising number of connected devices. Technical hurdles also include ensuring reliable data transmission among a variety of IoT devices. Security concerns are particularly pertinent; given constraints on cost, size, and power, IoT systems become more susceptible to security threats, resulting in significant financial and reputational risks [38,39]. Fig. 3 outlines multiple research challenges tied to IoT's exponential growth.

The proliferation of IoT devices has been remarkable, but the insufficient security measures in these devices have turned them into fertile ground for malicious activities. In the architecture of IoT, there is a four-layered design which is broadly acknowledged in the literature. These four layers are: the Perception layer, the Support layer, the Network layer, and the Application layer [40,41].

2.1. Perception layer

The perception (physical) layer is composed of sensors and actuators, which are integral for environmental data collection and subsequent actions [42]. These sensors, often referred to as nodes, are susceptible to various types of attacks, including node capturing, where an intruder might either seize or replace the node with a rogue one. Firmware or software updates can be exploited to inject malicious or false code, leading to code injection or false data injection attacks [43]. Additional vulnerabilities like side-channel attacks involving laser, power, or timing analyses are also prevalent in this layer [44]. Nodes are further exposed to eavesdropping attacks during data transmission, particularly when located in an open setting [45]. IoT devices often operate under power limitations, making them targets for attacks aimed at depleting their energy reserves. Boot-time attacks are also a concern since many IoT devices only enable security protocols post-booting.

2.2. Network layer

Data or network layer is responsible for transmitting data from the sensing components to computational elements, this layer is notably vulnerable to a range of attacks that affect multiple IoT devices [46]. Phishing attacks aim to compromise numerous devices, while attacks flood the network with counterfeit requests, often using IoT devices as botnets [47,48]. Routing attacks like Wormhole and Sinkhole alter the traffic paths by gaining unauthorized node access [49]. The layer is also at risk from Data Transit attacks during data transfers.

2.3. Support layer

Serving as an intermediary between the Network and Application

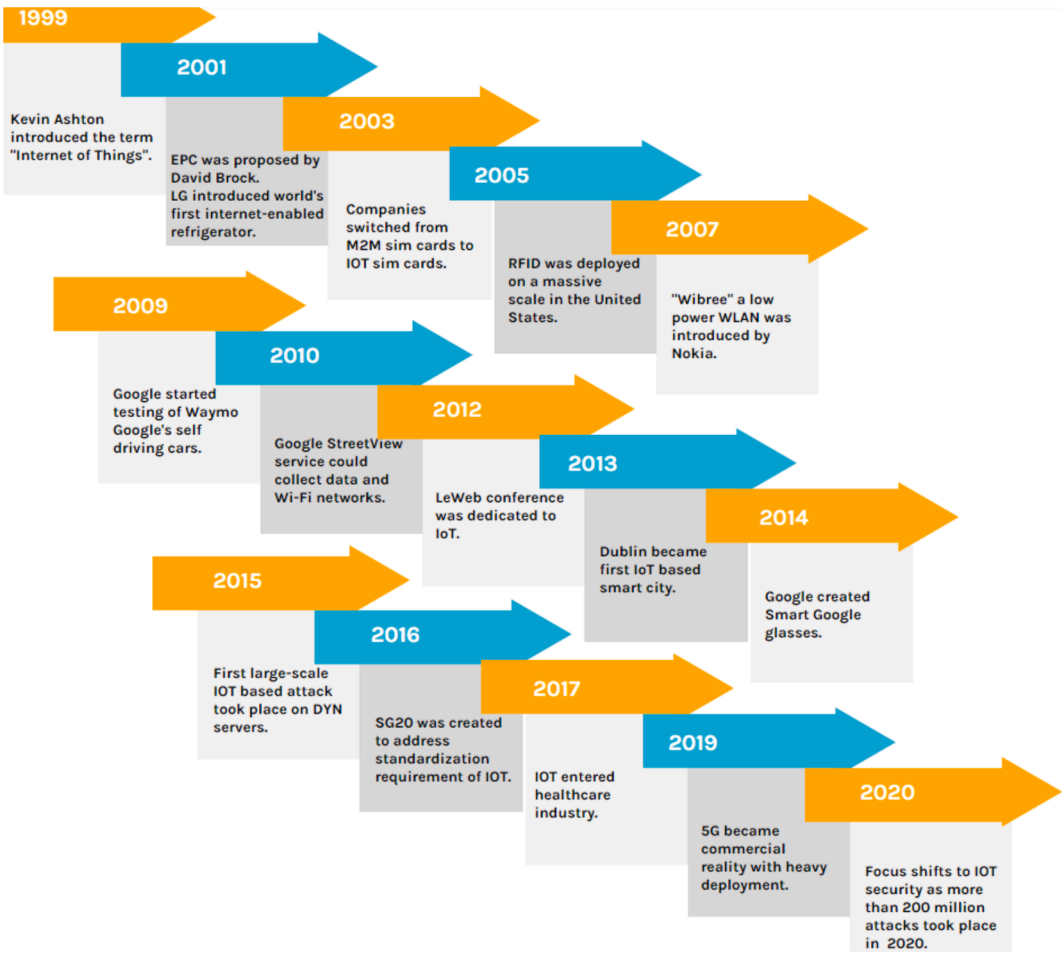


Fig. 2. Evolution of IoT.

layers, the support or control layer focuses on resource allocation, computation, and data storage. It is susceptible to various security threats including Distributed Denial of Service (DDoS), Man-in-the-

Middle (MITM), and Structured Query language (SQL) injection attacks [50]. Communication between clients and service providers often uses protocols such as the Message Queuing Telemetry Transport

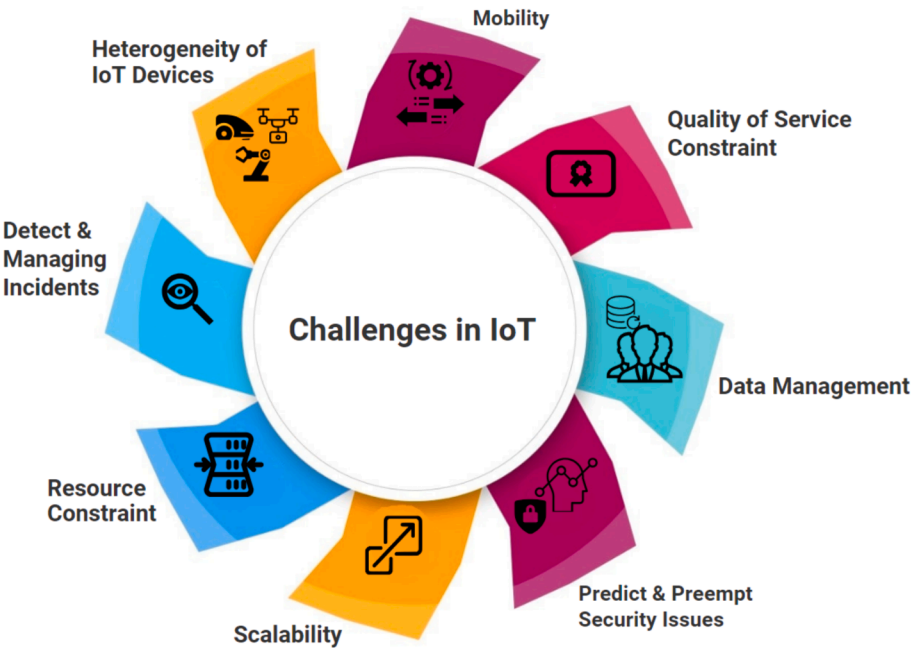


Fig. 3. IoT Research Challenges.

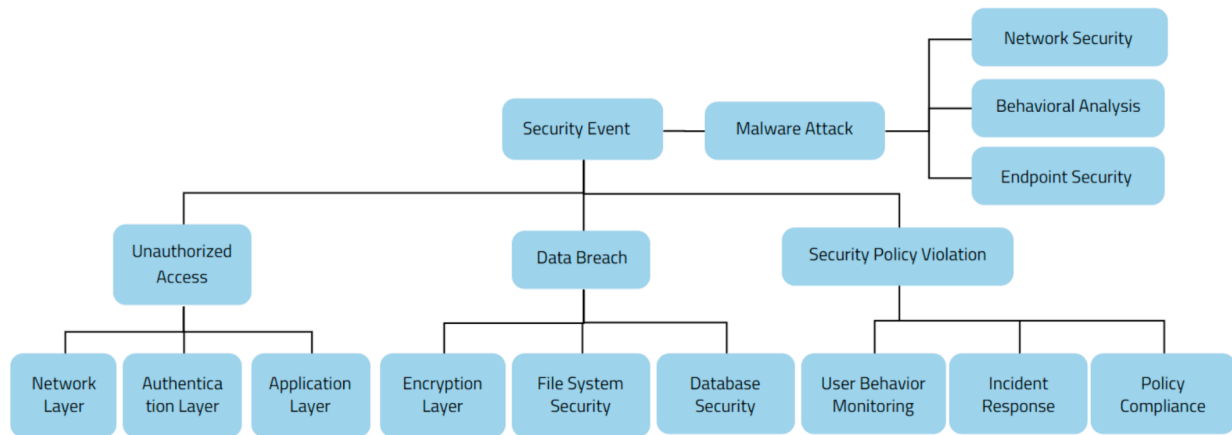


Fig. 4. Attack entry points for IoT devices.

(MQTT), which can be compromised in MITM attacks. Since the layer's primary function often involves data storage, database and cloud security are of paramount importance here.

2.4. Application layer

This layer houses intelligent applications for smart cities, homes, healthcare systems, and more. Given its direct interaction with end-users, data privacy and theft are major issues [51]. This layer is not immune to malicious code injections and experiences service interruption attacks, which are akin to Denial of Service (DoS) attacks. Access control attacks emerge as significant risks if unique user privileges are

compromised [52]. Sniffing attacks, facilitated by network traffic sniffing tools, pose a risk of confidential data being compromised [53].

In IoT systems, each layer presents unique vulnerabilities that can serve as entry points for cyberattacks. The Perception Layer is susceptible to sensor spoofing and physical tampering, while the Support Layer may suffer from unauthorized data access due to weak authentication protocols. The Network Layer is vulnerable to attacks such as MIMM and packet sniffing, which could disrupt communication. Finally, the Application Layer exposes risks related to web application vulnerabilities like injection and insecure applications. Therefore, each layer requires specific security measures to mitigate potential attacks. Fig. 4 shows attack entry points for the different layers of IoT devices.



Fig. 5. Developments in manufacturing provided by IoT technology.

3. IoT in manufacturing

IoT has emerged as a pivotal force in the manufacturing industry, often serving as the foundation of Industry 4.0 (I4.0). By incorporating smart sensors, machinery, and other connected equipment, IoT has enabled real-time monitoring and data analytics. This transformation has had a profound impact on various aspects of manufacturing, such as efficiency, safety, and overall decision-making. One of the most noteworthy applications of IoT in manufacturing is predictive maintenance. Sensors can identify the risk of equipment failures before they occur, thereby minimizing downtime and enhancing productivity. Similarly, process optimization is another crucial application. IoT devices can continuously collect data, which is then analyzed to fine-tune manufacturing processes. This level of insight contributes to reduced waste and greater operational efficiency. Moreover, quality control has seen significant improvements through the adoption of IoT. Cameras and sensors can quickly and accurately identify product defects, surpassing the speed and reliability of human inspection. Additionally, IoT has streamlined supply chain and inventory management by tracking materials throughout the supply chain, making it more transparent and efficient. Energy management also benefits from IoT devices, which can scrutinize energy usage and suggest areas for reduction. In terms of benefits, the cost-saving potential of IoT is substantial. Reduced machine downtime, lower energy costs, and streamlined operations all contribute to this saving. Beyond that, IoT-equipped manufacturers enjoy higher levels of productivity and can maintain a competitive edge in the market. The insights also facilitate more accurate business forecasting and data-driven decision-making. However, the adoption of IoT is not without its challenges. Security is a major concern as the increased connectivity opens up more avenues for potential cyber-attacks. Companies may also find themselves grappling with the massive volumes of data generated, leading to data overload. Moreover, integrating IoT into older manufacturing systems can be a complex, costly, and time-consuming process.

In the manufacturing landscape that is increasingly interconnected through digital means, cybersecurity has evolved into a cornerstone that may not be originally part of Lean manufacturing principles but complements them effectively. Lean manufacturing is committed to minimizing waste, a goal that coincides with cybersecurity's aim to prevent the waste of resources and time due to cyber-attacks. Handling the aftermath of a cyber incident can lead to loss of productive hours, the financial strain of system recovery, and the additional effort in managing reputational damage [54]. Cybersecurity adds value by safeguarding data, thereby bolstering the firm's reputation and enhancing customer trust and loyalty. Stable and predictable operations are also a hallmark of Lean manufacturing, an outcome that cybersecurity helps to ensure. By securing the underlying technological infrastructure, cybersecurity ensures uninterrupted production, thereby sustaining a consistent flow of products to the consumer market. In modern manufacturing, managing cybersecurity isn't just limited to securing contemporary Information Technology (IT) devices like user endpoints running on Windows or macOS. It also encompasses the challenge of protecting obsolete machinery that operates on outdated versions of software, which no longer receive updates or support. The manufacturing environment is also rife with Operational Technology (OT), and Big Data analytics serve as invaluable tools for real-time monitoring and assessment of network vulnerabilities and functions. These insights form a crucial part of an enterprise's overarching risk management strategy. To achieve effective security, an ideal approach involves integrating all these elements into an AI-based Network Intrusion Detection System (NIDS). Fig. 5 illustrates several improvements in manufacturing systems provided by IoT technology.

4. Taxonomy of IoT attacks

For many years, advancements in network technologies have

revolutionized data distribution and communication, paving the way for an array of new services. IoT has become a critical facilitator for enhanced communication, linking a myriad of devices to the Internet and harvesting data. This collected data aids companies in evaluating and predicting consumer habits, thereby refining their product quality. Modern applications of ML and DL are shaping network systems capable of advanced analytics and automation, offering user experiences that mimic human cognitive functions through embedded algorithms [55]. Moreover, the rise of cloud computing has induced significant shifts in networking paradigms. Users now have on-demand access to a multitude of applications, services, and computing and storage capabilities via the Internet. These developments offer a host of benefits, including adaptability, reduced administrative burdens, efficient resource utilization, and better reliability [56]. One of the most promising evolutions in network technology is the introduction of 5G networks. This fifth-generation mobile network standard is designed to connect virtually anything, ranging from machinery and objects to personal devices. 5G not only offers enhanced speed and connectivity but also introduces the concept of network slicing. This involves partitioning a single physical network into multiple virtual networks to better cater to the needs of diverse applications. With applications spanning from entertainment and online gaming to public safety and education, 5G is poised to offer faster download speeds, real-time responses, and consistent connectivity, enabling both enterprises and consumers to discover new technological frontiers [57]. With growing benefits comes a considerable set of novel security risks. This necessitates the ongoing creation of detection mechanisms and security countermeasures tailored to these evolving network technologies. A network attack refers to any action or series of actions aimed at compromising the integrity, confidentiality, or availability of a computer network and its associated resources. Network attacks can be perpetrated by individuals, groups, or entities with malicious intent, such as unauthorized access, data theft, service disruption, or resource manipulation. These attacks are often categorized into various types based on their target and method of execution.

4.1. The distributed Denial of service (DDoS)

One of the most severe threats in network security is the DDoS attacks. They are designed to overload a target system, effectively rendering its services unusable. IoT devices, given their proliferation and commonly unsecured nature, are particularly conducive for staging these attacks. Often, the device owners remain unaware that their devices, such as baby monitors or smart toys with restricted user interfaces, have been compromised and are part of a botnet [58]. As the number of IoT devices continues to soar, the imperative to identify and neutralize compromised devices in a timely manner has never been greater. High-profile service providers like Amazon Web Services (AWS) have fallen prey to such DDoS attacks, as have security firms like Krebs On Security and Cloudflare [59,60]. These incidents not only cause financial losses but also tarnish the reputations of the targeted organizations. In a DoS attack, the attacker overloads the target's resources by sending spurious requests, causing service disruption. DDoS attacks are essentially intensified DoS attacks involving multiple sources of fake requests, making them particularly difficult to counteract. Various forms of DDoS attacks exist, including but not limited to, Flood, Teardrop, Smurf, and Ping of Death attacks, often involving botnets for execution. Additionally, DDoS attacks can be categorized into Reflection and Amplification types. In Reflection attacks, the request and response sizes are equal, whereas, in Amplification attacks, the response size is significantly larger than the request [61,62]. The primary distinction between DDoS and DoS attacks lies in the number and variety of systems employed to initiate the attack. In a DDoS assault, a diverse array of systems—ranging from desktop computers and servers to IoT devices—are harnessed to bombard the target [63]. The initial phase of such an attack involves gaining control over these various systems, often by exploiting existing security weaknesses. In the context of IoT, multiple security

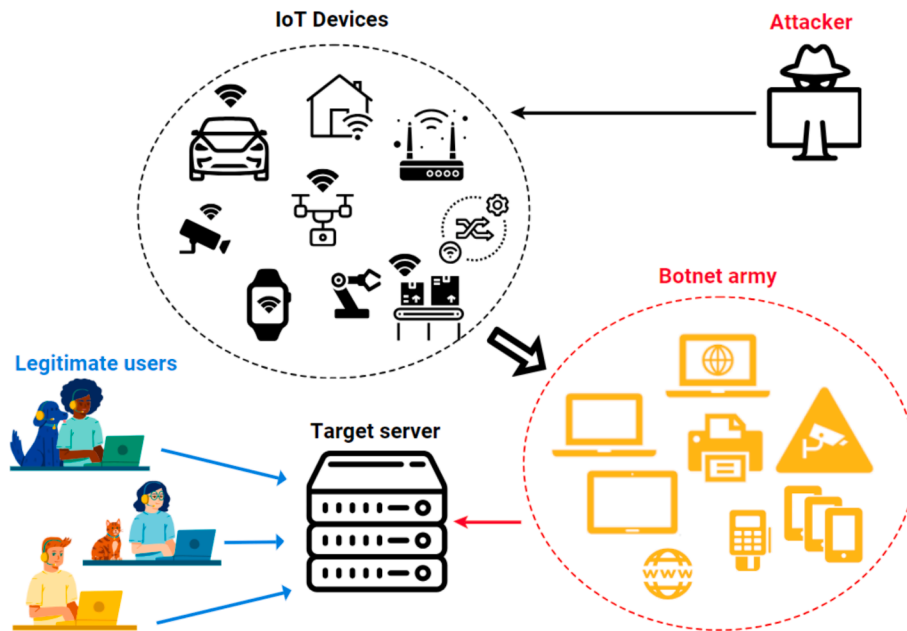


Fig. 6. The mechanism behind a DDoS attack

vulnerabilities serve as potential entry points for attackers, who are increasingly capitalizing on these flaws [64]. In the context of DDoS attacks, attackers deploy malicious scripts to identify and compromise vulnerable devices. This approach is notably effective against IoT devices due to their generally weaker security protocols [65]. After gaining control of these insecure devices, they are converted into bots, collectively forming what is known as a botnet. This assembled botnet can subsequently be mobilized by its controller to initiate a large-scale attack or can be sold for use by other malicious entities. DDoS attacks occur when this botnet is utilized to flood the target with an overwhelming volume of requests, thereby incapacitating it. Every server has a set capacity to manage a specific number of requests simultaneously. Attackers aim to surpass this limit by flooding the server with an excessive volume of requests, causing it to become overwhelmed and thereby cease functioning [88]. The vulnerabilities in IoT devices make them attractive targets for attackers. These vulnerabilities span software, hardware, and physical aspects. Moreover, IoT devices often employ protocols that are susceptible to DDoS attacks, specifically Constrained Application Protocol (CoAP), Routing Protocol for Low-Power and Lossy Networks (RPL), and Internet Protocol v6 over Low-Power Wireless Personal Area Networks (IPv6LoWPAN) [89]. Fig 6 attempts to depict the mechanism behind a DDoS attack.

A DL-based mechanism called DeepDefense was introduced for detecting DDoS attacks by analyzing the sequence patterns of network traffic [66]. The team utilized the UNB ISCX intrusion detection evaluation 2012 (ISCX2012) dataset, and employed the Recurrent Neural Network (RNN) algorithm to construct the model. They processed 20 network traffic attributes from ISCX2012 to generate a 3D feature map using a sliding time window approach. Two subsets of data from ISCX2012, named data14 and data15, contained 9.6 million and 34.9 million packets, respectively. The total training samples for these datasets were 15,176 and 233,450. DeepDefense showed a significant reduction in error rates; specifically, it decreased the error rate by 39.69 % in smaller datasets compared to traditional ML methods. For more extensive datasets, the reduction varied between 7.517 % and 2.103 %. For further research, the authors recommend enhancing the diversity of DDoS attack types and system configurations to more rigorously test the DeepDefense model, and also to compare it against other ML algorithms. Baek et al. [67]. conducted a separate study that focused on recognizing DDoS attacks affecting both network and service levels in the Bitcoin

ecosystem. Their dataset consisted of real DDoS attacks and included information such as the affected service, attack date, and attack type. The researchers utilized Principal Component Analysis (PCA) for feature extraction and employed Multilayer Perceptrons (MLP) for attack detection. They divided their dataset into training, validation, and testing sets in a 6:2:2 ratio. The achieved accuracy for identifying DDoS attacks was approximately 50 %, while the classification accuracy for normal block data was around 70 % with the epoch unit set to 100. In another research endeavor by Sabeel et al. [68] Deep Neural Networks (DNN) and Long Short-Term Memory (LSTM) networks were employed for the binary classification of unknown DoS and DDoS attacks. They used the CICIDS2017 dataset for training and a new dataset called ANTS2019 for testing. A total of 78 features were used for training, while 77 features were used for testing. The dataset was split into training and testing sets at a 75–25 % ratio. When trained with CICIDS2017 and a part of ANTS2019, DNN achieved an evaluation accuracy of 99.68 %. Upon retraining the models with new unknown attacks, true positive rates of 99.8 % and 99.9 % were achieved for DNN and LSTM, respectively. The authors concluded that the models should be periodically updated with new attack data to maintain high performance.

Elsayed and colleagues [69], developed an Intrusion Detection System (IDS) known as DDoSNet, which melds the capabilities of an Autoencoder (AE) and RNN. In their study, they utilized the recently released CICDDoS2019 dataset, comprising 80 flow features. They employed PCA for feature engineering, narrowing the input features down to 77. The datasets were partitioned into 161,523 training samples, 46,150 validation samples, and 23,000 testing samples. The model demonstrated a 99 % accuracy rate, outperforming other machine learning techniques such as Support Vector Machines (SVM), Decision Trees (DT), Naive Bayes (NB), Random Forests (RF), Boosters, and Logistic Regression (LR). Future research aims to validate the model's effectiveness across varied datasets and extend its functionality to multiclass classification, as the current model employs a binary classification scheme. Doriguzzi-Corin et al. [70] proposed a model leveraging Convolutional Neural Networks (CNN) to distinguish between benign and malicious traffic flows. They used datasets such as CICIDS2018, CICIDS2017, and ISCX2012, available through the Canadian Institute for Cybersecurity at the University of New Brunswick (UNB). Their selection comprised 37,378 benign and an equal number of

DDoS flows from the ISCX2012 dataset, 97,718 each from the CICIDS2017 dataset, and 360,832 each from the CICIDS2018 dataset. After preprocessing, each dataset was split into 90–10 % train-test sets. The model showed remarkable accuracy for each dataset—99.87 %, 99.67 %, and 98.88 %, respectively. When a unified dataset called UNB201X was formed by amalgamating data from all years, the model's accuracy reached 99.46 %. For future endeavors, the authors aim to refine the preprocessing tools and extend the dataset labels. Ahuja et al. [71] implemented a variety of DL algorithms, including CNN, RNN, LSTM, a hybrid of CNN and LSTM (CNN-LSTM), Support Vector Classifier-Self Organizing Map (SVC-SOM), and Stacked Autoencoder-Multi-Layer Perceptron (SAE-MLP) to identify DDoS attacks. Their dataset contained a total of 22 features. They employed two different optimization algorithms: Stochastic Gradient Descent (SGD) for the initial 10 epochs and the Adam stochastic optimization for the succeeding 150 epochs. The models were evaluated on multiple metrics such as accuracy, precision, recall, F-score. The SAE-MLP algorithm achieved the highest classification accuracy of 99.75 % [72].

A study investigated the application of DL in the identification of DDoS attacks through both packet-based and period-based techniques [73]. They introduced a DL model named DeepDDoS, utilizing Spark as its big data processing platform. For feature selection, the team relied on maximal information coefficient and mutual information. LSTM was chosen for the training stage due to its efficacy in handling longer sequences. The primary aim was to isolate anomalous network flow using minimal computational resources. The CICIDS2017 dataset, was employed for experimentation. The model boasted an accuracy rate exceeding 99 % when it processed five consecutive packets. Quamar Niyaz et al. [74] developed a DL model specifically designed to identify multi-vector DDoS attacks in software-defined networks. They employed a Stacked Autoencoder (SAE) and gathered data from both a public network connected to the internet and a private lab network with DDoS attacks. Max-min normalization was applied to the separated training and testing datasets. For comparative purposes, models based on soft-max and traditional Neural Networks (NN) were also constructed. The SAE-based model outperformed these, achieving an accuracy of 95.65 %. Future research by the team aims to construct a Network Intrusion Detection System (NIDS) capable of detecting various types of attacks and to further explore DL techniques for feature extraction. Pande et al. [75] aimed to construct a ML-based model for the detection of DDoS attacks. They utilized the ping of death technique to simulate a DDoS attack and deployed an RF model for its detection. They utilized the NSL-KDD dataset, which included a training set of 125,973 records and a testing set comprising 22,544 instances across 41 attributes. The model took 8.71 s to build and 1.28 s to test, yielding an accuracy of 99.76 %. Future work intends to integrate deep learning techniques for improved classification. Radivilova et al. [76] embarked on a study to assess various strategies for DDoS attack detection via network traffic, using the SNMP-MIB dataset. RF was utilized for classification. Feature extraction involved recurrence analysis. The key metric employed was accuracy. Their findings indicated that early detection is feasible when the average attack volume constitutes 15–20 % of the typical network traffic patterns. Filho et al. [77] proposed an intelligent DDoS detection system leveraging ML algorithms, with the objective to identify both high and low-volume attacks. RF was selected for its high precision across 28 variables. Evaluation metrics included recall, precision, and the F1 score. The study made use of four benchmark datasets for intrusion detection, namely, CICIDS2017, CIC-DoS2017, CICIDS2018, and customized sets. The model displayed over 93 % precision and detection rate, while maintaining a false alarm rate below 1 %. Similarly, Vijayanand et al. [78] introduced a system designed to pinpoint novel DoS attacks. They employed a series of DL algorithms arranged in a hierarchical fashion to accurately assess smart meter network traffic. The proposed approach tackles challenges associated with large data volumes and complex feature sets. Evaluation metrics were based on 9,919 records from the CICIDS2017 dataset. The

system's effectiveness was gauged against standard MLP algorithm and hierarchical SVM algorithm, achieving efficiency rates of 39.78 % and 99.99 %, respectively.

Robinson and Thomas [79] assessed the efficiency of various supervised algorithms using the CAIDA DoS attack dataset. Their study revealed that nearly all ML algorithms attained over 99 % accuracy on the CAIDA Conficker dataset, which featured easily distinguishable attack vectors generated from large botnets. Barati et al. [80] examined the CAIDA dataset and developed a hybrid ML model for DDoS attack identification. Genetic Algorithms (GA) and Artificial Neural Networks (ANN) were employed for feature selection and attack detection, respectively. The model exhibited an Area Under the Curve (AUC) of 99.91 %, indicating excellent performance. Future research will focus on testing the model's robustness across different datasets. Kim et al. [81] constructed a CNN-based model for detecting DoS attacks. They utilized the KDD-99 and CICIDS2018 datasets. The performance of this model surpassed that of a RNN model in identifying specific DoS attacks with similar attributes, achieving accuracies of 99 % or higher in both binary and multiclass classifications. Additionally, Rehman et al. [82] developed a method using Gated Recurrent Units (GRU) for DDoS attack detection. Their high-efficiency approach, dubbed DIDDOS, applied multiple classification algorithms including GRU, RNN, NB, and Sequential Minimal Optimization (SMO) on the CICDDoS2019 dataset. Notably, the GRU model exhibited exceptional accuracy rates, reaching 99.69 % for reflection attacks and 99.94 % for exploitation attacks.

4.2. Phishing attacks

Several research efforts have been devoted to creating and evaluating models for phishing attack detection. For example, Alam et al. [83] aimed to thwart phishing attacks by crafting a detection model that utilized RF and DT. They made use of a standard Kaggle dataset with 32 features related to phishing attacks and employed PCA for feature selection. The RF algorithm achieved an impressive accuracy rate of 97 %, and it also managed to mitigate the issue of overfitting. Mohammad et al. [84] designed a Self-Structuring Artificial Neural Network (SSANN) for recognizing phishing websites. The network dynamically adapts its structure to effectively detect highly variable web pages. The model demonstrated a high tolerance for noisy data, fault tolerance, and considerable predictive accuracy. By manipulating the learning rate and adding more neurons to the hidden layer, the network was optimized for generalizability. Training, validation, and testing accuracies were 94.07 %, 91.31 %, and 92.18 % across 1000 epochs, respectively. Thabtah et al. [85] proposed an alternative to the time-consuming trial and error method commonly used in NN training. They designed a dynamic, self-structuring anti-phishing NN that adjusts its parameters during the training phase to create accurate, non-overfitting models. Their approach was tested on a dataset of over 11,000 websites from the UCI repository. The model demonstrated superior predictive accuracy compared to Bayesian Networks (BN) and DT, reaching an average accuracy of 93.06 % when pre-processed using information gain. Rendall et al. [86] introduced a two-layer detection framework for identifying phishing websites. They employed four models: MLP, SVM, NB, and DT. Their approach involved multiple classification rounds for phishing domains, with additional classifications triggered if the system's confidence level fell below a certain threshold. The model, evaluated on a custom dataset containing 5,995 phishing and 7,053 benign records, achieved its highest accuracy of 86 % using MLP and DT. Li et al. [87] developed a stacking model that integrated gradient boost algorithms with DT for real-time phishing webpage detection. They utilized two custom datasets: 50 K-PD with approximately 49,947 samples and 50 K-IPD with around 53,103 samples. The model achieved 97.30 % and 98.60 % accuracy on these datasets, respectively. Batnaru et al. [88] explored phishing detection by using supervised machine learning models and custom features extracted from Uniform Resource Locator (URL). They used a combined dataset from Kaggle and PhishTank,

Table 2
Classification of attacks (Continued).

	Attack Mechanism			Attack Type	IoT Attack Layer			Delivery Mechanism			Attack Goal				Attack Size		Security Hole														
	Resource Exhaustion Attacks	Network Attacks	Unauthorized Access Attacks	Data Manipulation and Theft	Reconnaissance Attacks	Sleep and Boot Attacks	Volumetric-Based Attacks	Protocol-Based Attacks	Application-Based Attacks	Perception Layer	Support Layer	Network Layer	Application Layer	Packets	Malicious Code	Botnet Tools	Exploits	Targeting User Data or Credentials	Consuming the Bandwidth of Target and Internet	Consuming Resources of Server and Firewall	Exhausting Target Resources	Targeting or Collecting Data	Hijacking	Bits per Second (bps)	Packets per Second (pps)	Requests per Second (rps)	Unsecure Connection	Weak Password	Out of Date Firmware	Software Vulnerabilities	Data Handling Vulnerabilities
Access control attack			x						x				x								x										x
Account Lockout			x					x			x							x					x					x			
Account Takeover			x					x					x					x					x					x			
Active Reconnaissance					x			x					x									x									
API Abuse									x				x							x						x					
Backdoor			x					x					x		x								x		x					x	
Booting vulnerabilities						x		x				x					x				x				x				x		
Botnets								x					x			x			x						x				x		
Credential Stuffing				x					x				x					x					x			x		x			
XSS				x					x				x		x					x						x				x	
Data transit attack				x				x					x									x									
DDoS	x						x					x		x					x					x				x			
DoS	x						x					x		x					x					x				x			
Eavesdropping				x				x					x					x					x								x
Exploit attack			x					x					x				x			x						x				x	
False data injection attack				x				x					x		x					x						x				x	
Flooding	x						x					x		x					x					x			x				
Jamming attack		x					x			x				x					x					x							
Malicious code injection attack				x					x				x		x				x							x				x	

	Attack Mechanism			Attack Type	IoT Attack Layer			Delivery Mechanism			Attack Goal				Attack Size		Security Hole														
	Resource Exhaustion Attacks	Network Attacks	Unauthorized Access Attacks	Data Manipulation and Theft	Reconnaissance Attacks	Sleep and Boot Attacks	Volumetric-Based Attacks	Protocol-Based Attacks	Application-Based Attacks	Perception Layer	Support Layer	Network Layer	Application Layer	Packets	Malicious Code	Botnet Tools	Exploits	Targeting User Data or Credentials	Consuming the Bandwidth of Target and Internet	Consuming Resources of Server and Firewall	Exhausting Target Resources	Targeting or Collecting Data	Hijacking	Bits per Second (bps)	Packets per Second (pps)	Requests per Second (rps)	Unsecure Connection	Weak Password	Out of Date Firmware	Software Vulnerabilities	Data Handling Vulnerabilities
Man in the Middle									x			x						x				x					x				
Node Capture Attack			x						x								x				x				x				x		
Passive Reconnaissance					x				x				x								x					x					
Phishing attack				x					x									x					x								x
Ping of Death attacks		x					x					x		x					x					x							
Ransomware									x				x									x									
Reprogram attack			x						x				x									x								x	
Routing attack		x					x				x						x				x					x					x
Service interruption attack	x						x						x							x					x						x
Session hijacking				x					x				x										x								
Side channel attack				x					x				x										x								
Sleep deprivation attack						x		x		x											x					x		x			
Smurf		x					x					x		x					x						x						
Sniffing				x					x				x					x					x								x
Storage attack									x		x											x					x				x
Teardrop		x							x				x	x					x						x					x	
Traffic Analysis				x					x			x											x								
Vulnerability Scanning					x				x				x				x						x								x
War Driving						x			x														x								
Zero access			x						x			x				x					x								x		

XSS: Cross Site Scripting, API: Application Programming Interface

containing 100,000 URLs. The RF model was selected as the most effective based on F1 scores. An unbalanced dataset, comprising 305,737 benign URLs and 74,436 phishing URLs, was used for the model evaluation, achieving an accuracy of 99.29 %. The model outperformed

Google Safe Browsing (GSB) protection mechanism, and future research will look into its applicability to other datasets and its resilience to adversarial attacks. Rao et al. [89] proposed PhishDump, a mobile application that utilized LSTM and SVM algorithms to identify authentic

and fraudulent websites on mobile platforms. Data was collected from Alexa, OpenPhish, and PhishTank. The app was effective, classifying malicious websites in just 621 ms and achieving an accuracy of 97.30 %. However, the model has limitations, including susceptibility to manipulated URLs. Marchal et al. [90] provided a comprehensive review of phishing attack detection strategies, offering guidelines on designing and assessing detection techniques. They discussed various design trade-offs and suggested metrics for evaluating detection methods, encouraging the academic community to focus on real-world applications. Das et al. [91] revisited existing research on phishing and spear phishing across various security domains such as real-time detection and dataset quality. They highlighted the challenges and reviewed current solutions, aiming to pave the way for the development of more resilient anti-phishing methods.

4.3. Malware attack

Barut et al. [92] set out to evaluate the performance of ML algorithms in detecting encrypted malware data. The researchers generated two datasets: the first one using Stratosphere IPS, which comprised 20 different malware classes, and the second one utilizing CICIDS2017. They employed chi-square tests during feature engineering to select 200 flow features. RF was identified as the most effective algorithm, boasting a detection rate of 99.996 %. Marin et al. [93] introduced a model called DeepMAL for detecting malware in encrypted network traffic. They used the USTCTFC2016 dataset, which contained labeled traffic as either normal or malicious, and included 10 malware types. The researchers experimented with two data representations: packet-based and flow-based. The latter yielded better results, with DeepMAL achieving a 99.9 % detection rate for the Rbot botnet and surpassing the performance of RF for Neris and Virut botnets. Park et al. [94] examined the effectiveness of the RF algorithm in identifying various cyber attack types, using the Kyoto 2006 + dataset. The dataset included traffic data collected between November 2006 and December 2015, and was divided into three classes: normal, attack, and shellcode. For the study, the researchers chose data from May 2014 for training and data from April 2014 for testing. They selected 17 features and normalized the data before running the experiments. The model achieved an F-Score of 99 %, though the performance varied for different types of attacks. David et al. [95] implemented a DL model using a Deep Belief Network (DBN) algorithm to automatically identify and classify new variants of malware. The researchers compiled their own dataset comprising 1800 samples divided across six malware types: Zeus, Carberp, Spy-Eye, Cidox, Andromeda, and DarkCome, each containing 300 variants. The DBN architecture included eight layers, with the final layer housing 30 neurons. The training was unsupervised, employing 1200 vectors for training and 600 for testing. Noise ratio was set at 0.2 and training epochs at 1000 to denoise the AE. The resulting model achieved an impressive accuracy rate of 98.6 %. Wu et al. [96] employed reinforcement learning to emulate potential attackers by generating new malware samples, providing defensive teams with realistic attack scenarios. They advanced the existing gym-malware model into gym-plus, which expanded the action space by adding new activities and allowed for the alteration of malicious portable executable files. Utilizing the public EMBER dataset, the model underwent retraining to enhance the detection rate. Algorithms such as Deep Q-Network (DQN), State-Action-Reward-State-Action (SARSA), and Double DQN were implemented, with DQN outperforming the others in policy formulation. After retraining, malware detection accuracy saw a significant jump from 15.75 % to 93.5 %. Letteri et al. [97] investigated the MTA KDD 19 dataset while applying dataset optimization techniques for malware traffic detection. They employed two dataset optimization methods: one involving dimensionality reduction using optimized AE, and the other a rank relevance weight-based feature selection method. In the latter, feature selection was done in two stages, starting with 5-fold cross-validation for dataset tampering followed by backward feature

Table 3
Definition of each attack.

Attack Type	Definition
DoS	Targets availability of services by overwhelming a system with excessive requests.
DDoS	Similar to DoS but involves multiple systems attacking a single target simultaneously.
Backdoor	Malicious software allowing unauthorized remote access to a system.
Teardrop	Exploits vulnerabilities in TCP/IP fragmentation, causing crashes or reboots.
Flooding	Sends a flood of traffic to overwhelm a network or service, a type of DoS.
Smurf	Amplifies a network request to flood a target with responses.
Ping of Death	Sends malformed or oversized pings to crash systems.
Botnets	Networks of compromised systems used to launch coordinated attacks. For example, Mirai and Gafgyt.
Node Capture Attack	Physically compromising a network node to control or manipulate it.
Malicious Code Injection Attack	Inserts or "injects" harmful code into a system.
False Data Injection Attack	Injects inaccurate data to deceive or mislead a system.
Side Channel Attack	Gains information from the physical implementation of a system.
Eavesdropping	Secretly intercepts and listens to data passing over a network.
Sleep Deprivation Attack	Prevents a system from entering low-power states, draining its energy.
Bootling Vulnerabilities	Exploits flaws in the booting process to compromise a system.
Man in the Middle	Intercepts and potentially alters communication between two parties.
Data Transit Attack	Targets data as it moves between systems or locations.
Phishing Attack	Uses deception to trick users into revealing sensitive information.
Routing Attack	Manipulates routing tables to redirect or block data flow.
Storage Attack	Targets storage systems to corrupt or steal data.
Exploit Attack	Utilizes known vulnerabilities to compromise a system.
Sniffing Attack	Captures and analyzes network traffic for data extraction.
Access Control Attack	Bypasses or compromises mechanisms that limit system access.
Service Interruption Attack	Causes service downtime, affecting availability.
Jamming Attack	Interferes with the physical layer of wireless communication.
Reprogram Attack	Alters the firmware or software on a device.
XSS	Injects malicious scripts into web pages viewed by others.
Session Hijacking	Takes over a user's session to gain unauthorized access.
Ransomware	Encrypts data and demands payment for its release.
Active Reconnaissance	Gathers information through direct interaction with the target.
Passive Reconnaissance	Collects information without directly interacting with the target.
Traffic Analysis	Studies the flow and nature of network traffic for malicious activity.
War Driving	Locates wireless networks by driving around an area.
Zero Access	A rootkit that creates a hidden platform for other malicious activities.
Credential Stuffing	Automated attacks using stolen credentials to gain unauthorized access.
Account Takeover	Gains control of a user account through malicious means.
Account Lockout	Deliberately locks users out of their accounts, often by flooding login attempts.
Vulnerability Scanning	Searches for weaknesses in a system or network.
API Abuse	Exploits vulnerabilities in an Application Programming Interface.

TCP: Transmission Control Protocol, IP: Internet Protocol

elimination. For the optimized AE, the NN had 33 input and output neurons with an 85–15 % train-validation split, and the training set was further divided to allocate 15 % for testing. The rank relevance weight-based approach yielded the highest accuracy, achieving a rate of 99.60 %.

Table 4

Grouping rules used to classify the listed attacks.

Rule	Description
Unsecure connection	When connected devices are left unprotected by firewalls, it becomes significantly easier for attackers to compromise these devices and launch various types of cyber-attacks.
Weak passwords	Attackers often utilize brute-force techniques to crack weak passwords, a situation exacerbated in the IoT landscape where companies frequently employ identical default passwords across multiple devices. When attackers compromise one device, they often use that as a springboard to target additional devices from the same manufacturer.
Out of date firmware	Many IoT devices are susceptible to attacks due to their use of outdated or insecure firmware. This lack of secure firmware updates leaves the devices vulnerable to cyber threats.
Software vulnerabilities	Skilled attackers closely monitor software vulnerabilities, exploiting them before a security patch becomes available from the manufacturer. Such software weaknesses are challenging to discover and leverage but provide a fruitful avenue for cyber-attacks.
Data Handling Vulnerabilities	As connected devices frequently send and receive data, any vulnerabilities in the data transmission process—specially to cloud services or servers—can be readily exploited by malicious actors.
Volumetric-Based Attack	This type of attack seeks to overwhelm a network's bandwidth by flooding it with a high volume of data packets. The primary goal is to saturate the target's internet connection, making it inaccessible to legitimate users. Examples include UDP flood attacks and ICMP flood attacks.
Protocol-Based Attack	These attacks exploit vulnerabilities in network protocols to disrupt the normal functioning of the target. They often consume resources by manipulating session and handshake protocols. Examples include SYN/ACK attacks and Ping of Death attacks.
Application-Layer-Based Attack	This attack targets the top layer of the OSI model where web servers operate. Attackers focus on specific application vulnerabilities to exhaust server resources, thereby disrupting the application's functionality. Examples include HTTP GET/POST attacks and Slowloris attacks.
UDP: User Datagram Protocol, ICMP: Internet Control Message Protocol, SYN/ACK: Synchronize and Acknowledge, OSI: Open Systems Interconnection	

4.4. Botnet attacks

Ramos et al. [98] designed an innovative approach for botnet traffic detection by leveraging supervised learning algorithms and an enhanced dataset. They evaluated the performance of five ML classifiers: DT, RF, SVM, NB, and K-Nearest Neighbors (KNN), on two distinct datasets, CICIDS2018 and ISOT HTTP Botnet. For both datasets, an analysis of network flow metrics and feature selection was performed. The ISOT dataset ended up with 20 selected attributes like source and destination port numbers and transfer protocols, while CICIDS2018 had 19 similar types of attributes. Five-fold cross-validation was employed, and 80 % of the botnet samples were utilized for training purposes, with the remainder for testing. In the CICIDS2018 dataset, RF and DT yielded the highest accuracy rates, both scoring 99.99 %. For the ISOT HTTP dataset, RF and DT again were the top performers, achieving accuracy rates of 99.94 % and 99.90 % respectively. In a related study, Pektas and Akerman [99] employed DL techniques in conjunction with flow-based methods to detect botnet activity. They used two datasets for their research, namely CTU-13 and ISOT HTTP, which contained both benign and malicious data. The study integrated two DL algorithms, MLP and LSTM, for analysis. The ISOT dataset included two botnet types, Waledac and Zeus, whereas CTU-13 contained seven botnet families. For the ISOT dataset, the implemented approach attained an F-score of 98.8 %, while for the CTU-13 dataset, the F-score reached 99.1 %.

4.5. Zero-Day attacks

Beaver et al. [100] carried out a study using ML techniques capable of differentiating between benign and harmful network traffic. Their research framework deployed Gradient Boosting (GB) algorithm coupled with DT to classify types of network traffic based on the KDD-99 dataset. The constructed model was effective in detecting 82 % of previously undetected attacks and recognized 89 % of untrained attacks. It achieved a detection rate of 94 %. Ahmed et.al. [101] introduced a DL model to identify zero-day botnet attacks in real-time. Their model leveraged a feed-forward backpropagation ANN and DNN algorithms. For reliable performance, a robust dataset is essential; thus, they utilized the CTU13 dataset sourced from the Botnet Capture Facility. This dataset had nine input features and consisted of 10,000 randomly selected flow records. The data was initially normalized, and Adam's optimizer was applied during model training. They adopted an 80–20 % train-test split. The model demonstrated remarkable performance, achieving accuracies greater than 99.6 % after 300 epochs. It also surpassed the performance metrics of NB, SVM, and traditional back-propagation algorithms. Future work by the researchers will investigate the efficacy of their proposed model across diverse datasets. An expanded list of different attacks is shown in Table 2 where an attempt has been made to classify different types of attacks based on different aspects, Table 3 shows the definition for each of these attacks, and Table 4 discusses the classification criteria that was used. The following references ([102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120,121,122,123,124,125]) were used to extract some of the information presented in Tables 2 - 4.

Each kind of cyber-attack comes with its own set of defining features. For instance, the attribute for a DoS attack could be the rate of data transfer to the targeted node, while for an injection attack, it might be the characteristics of the injected signal. Layers react differently to variations in these attributes. As an example, a DoS attack with moderate traffic might have a minimal effect on a controller operating a Real-Time Operating System (RTOS) that prioritizes control functions over communication tasks. Hence, for publicly shared datasets to be of value for broader research initiatives and to be utilized in training ML and DL models, they must encompass a wide range of attacks and associated attributes. Creating such datasets presents a challenge, as the number of attack types to be included could grow exponentially [126].

5. Threat detection systems

As we transition into an era where nearly every device utilized by humans is internet-connected, ensuring their security is of critical importance. Systems for network defense aim to fulfill three core objectives: confidentiality, availability, and integrity. Techniques for detecting and preventing network intrusions can be broadly categorized based on their focus: identification of threats, neutralization of these threats, or a combination of both. The literature identifies two primary mechanisms to thwart attacks: Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). IDS operates as a warning system, flagging potential intrusions without taking corrective action, while IPS proactively takes measures to counter detected threats [127]. However, IPS faces challenges in case of false positives, which may result in blocking legitimate users. IDS is often emphasized due to concerns regarding false alarm rates, especially in the realm of malware detection. Classification can further be done based on the target location of the intrusion detection: Host-based IDS, Network-based IDS (NIDS), or Hybrid IDS. Host-based IDS is tailored for individual systems, offering strong detection of internal intruders and thorough assessments of compromise extent, but it is costly due to its one-to-one requirement [128]. NIDS excels at detecting external threats and offers a protective umbrella for multiple hosts, although it struggles with high-volume traffic analysis in some cases [129]. Hybrid IDS combines the strengths of both host and network-based systems, offering increased

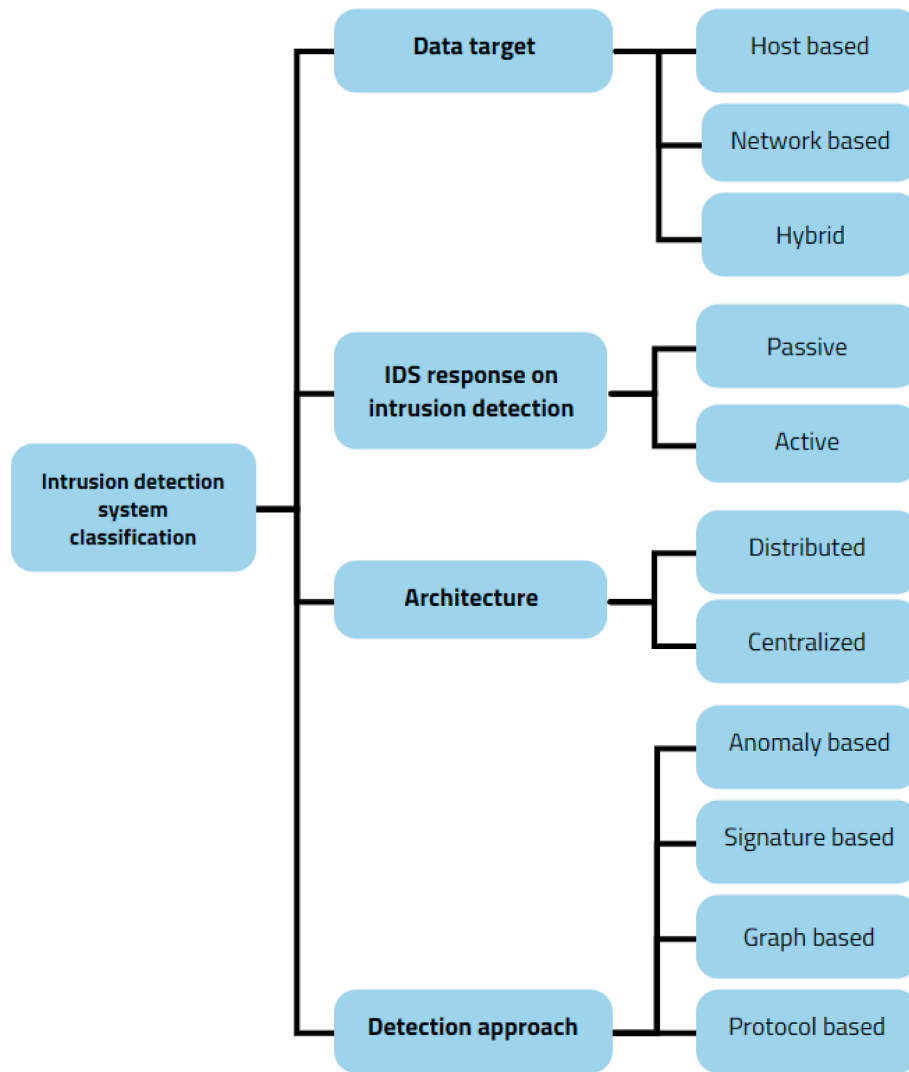


Fig. 7. Illustration of the different IDS.

security [130]. In terms of response mechanism, Active IDS undertake specific actions in response to certain alerts, whereas Passive IDS merely raises alarms or generates reports. From an architectural standpoint, Centralized IDS deploy individual monitoring units for each host, but lack scalability and are vulnerable to single points of failure. Conversely, Distributed IDS adopt a Peer-to-Peer (P2P) architecture where each monitoring unit also serves as an analysis unit, offering a more flexible and robust solution.

Threat detection predominantly employs two main strategies: Signature-based Detection and Anomaly-based Detection. Signature-based methods [131] encounter limitations particularly when dealing with Botnets, as these botnets frequently undergo mutations that alter their identifying signatures. This makes the approach less effective for recognizing new Botnet variants in real-world scenarios. In contrast, Anomaly-based detection techniques [132] gain favor because they operate on the assumption that the behavioral patterns of Botnet traffic will deviate from normal network traffic. Additional methodologies exist such as Community-Based Anomaly Detection [133], which employs Communication Graphs to identify Bots. However, this strategy requires a complete graph for accurate results. Some research employs specific protocols or structures for Botnet detection, but these approaches are not universally applicable if different Botnets use different structures [134]. Another tactic, known as the Bad Neighborhood method, is commonly used in Phishing Detection. It involves identifying

clusters of malicious IP addresses active over a specific duration. However, this method practicality is limited due to the widespread nature of DDoS attacks and the complications in forming such clusters [135,136]. Fig. 7 shows a tree diagram of different IDS and their relationship to each other while Table 5 shows a summary of the threat detection techniques. This paper will utilize an array of different ML and DL models to evaluate the fidelity and efficacy of an anomaly-based NIDS. In creating such intelligent systems, designers have to carefully select algorithms that are best suited for their specific area. For instance, RF is commonly employed due to its ensemble learning capabilities, which may contribute to building more adaptive systems. RF also excels in handling both categorical and continuous features, as well as missing data. However, its computational complexity poses challenges. SVM and NB follow as alternative choices, each with its set of pros and cons. DL models like DNN and RNN, along with ML models like RF and DT, have achieved accuracy rates of up to 100 %. Emerging research focuses on hybrid or ensemble models, incorporating advanced techniques such as CNN and LSTM for improved accuracy. Other models such as GRU could also be increasingly applied for their efficiency advantages.

The rise of IoT has been transformative for technological advancements. However, the future of IoT is heavily contingent on robust security protocols, a concern that is primarily addressed through the cybersecurity CIA model, encompassing Confidentiality, Integrity, and Availability. Attackers often exploit weak communication protocols,

Table 5
Categories of Threat Detection Techniques.

Method	Models	Examples	Implementation	Pros	Cons
Anomaly Detection Based Models	ML and DL [137]	Classification, Clustering, Detection, NN	Models are trained on datasets and then tested	Quick, Versatile, Easy to improve	Long time to train, requires large dataset to perform efficiently
Normal Traffic Behavior Based Model	Statistical [138]	Univariate, Multivariate, Time-Series, Markov Process	Historical data used to create mathematical model for normal traffic behavior	Simplicity	Statistical models tend to have low accuracy
	Payload Based Models [139]	N-Gram Analysis	Historical data used to create a model that learns the characteristics of normal traffic behavior and any deviation from the behavior is considered a threat	Simplicity	Slower response times, Very low versatility,
	Rules Based Model [140]	Fuzzy Rule, Association Rule	Normal traffic data creates rules and when these rules are broken then a threat is being detected	Simplicity	Requires longer time to generate the rules for the model, Low accuracy
Other Detection Models	Protocol Monitoring Models [141]	Communication and Application Protocols	Monitoring protocols at different network layers using computational models to detect threats	High accuracy for a specific attack at a specific layer	Low versatility
	Signal Processing Models [142]	Wavelet Analysis	Digital signals processing algorithms are used to analyze abnormal traffic patterns	High accuracy for a specific traffic analysis pattern	Low versatility

negatively impacting these three elements and tarnishing the reputation of service providers.

IoT devices produce varying scales of data, ranging from mere bytes to kilobytes per second depending on the application. The data can often be of critical importance, such as healthcare records or defense information. DDoS attacks stand out as prominent cybersecurity threats, and IoT devices are particularly vulnerable to being used as vectors for such attacks. This is because many IoT devices, like baby monitors or smart toys, have limited user interfaces, making it difficult for users to realize that their device has been compromised. As IoT has become more integrated into different industrial and household applications, the quest for robust security measures has gained prominence. Abu Al-Haija et al. [143] designed a DL based intelligent detection system for IoT cybersecurity, using CNN. They relied on the NSL-KDD dataset to validate their system, achieving a high accuracy rate exceeding 99.3 % for binary classification and 98.2 % for multiclass identification. Their future work

aims to expand their system to capture and analyze data packets within the IoT ecosystem. Khan et al. [144] put forward a blockchain-based security solution for IoT, harnessing the capabilities of Extreme Learning Machine (ELM). Their approach validated the integrity, confidentiality, and availability of blockchain-based smart homes. Preliminary data revealed minimal ELM overheads compared to the resulting cybersecurity benefits. They achieved an accuracy of 93.91 % using the NSL-KDD dataset, with future work centered on exploring diverse architectures and datasets. Ullah et al. [145] used a DL methodology to identify malware and pirated software across IoT networks. Their dataset originated from Google Code Jam (GCJ) [146], and their approach utilized TensorFlow for pirated feature detection and CNN for malware identification. The methodology outperformed existing techniques in IoT cybersecurity, and future work will focus on the identification of new malware families. Tama and Rhee [147] deployed DNN for classifying attacks and detecting anomalies in IoT networks. They used multiple datasets like CIDDs-001, UNSW-NB15, GPRS-WEP, and

Table 6
Qualities needed in IoT Datasets.

Quality	Description
Labeling	In a dataset, multiple records can correspond to a single cyber-attack. Therefore, each attack should be assigned a unique identifier that serves as a label for all records related to it. This is in addition to traditional labeling methods such as binary (normal/attack) or multiclass labeling. Utilizing unique identifiers for attacks enables more advanced detection and prevention systems that can correlate data. Automation is crucial for this labeling process as doing it manually would be overwhelming. Additionally, each record should indicate the system's operational state, either normal or failure.
Imbalance	A common issue in IoT Networks datasets is class imbalance. This problem could be addressed either through under-sampling and over-sampling methods, or by injecting a greater number of attacks during dataset generation.
Redundancy	Records that represent the same state don't offer new information. A dataset covering a week of standard operation could feasibly be condensed to just a few hours without losing crucial data. Eliminating redundant data is essential to conserve computational resources during the training of detection systems.
Synchronization	Synchronized timestamps in dataset records are vital to establish the causal relationships between system events and attacks. Asynchronous system nodes can produce out-of-order records, disrupting the accurate modeling of system behavior, an aspect especially crucial for synthesized datasets.
Scale	Most available datasets come from small-scale IoT devices. Real-world systems, however, may consist of thousands of interconnected sensors, actuators, and controllers, generating data that varies in scale and variable correlation compared to testbed data. As such, there's no guarantee that models trained on smaller datasets will scale effectively.

Table 7
List of available datasets.

Datasets	Availability
DARPA-1998	[161]
KDD-99	[162]
NSL-KDD	[163]
UNSW-NB15	[164]
CICIDS-2018	[165]
CTU-13	[166]
IoTID 20	[167]
Kyoto 2006+	[168]
CERT v6	[169]
CERT v4	[170]
SNMP-MIB	[171]
ISCX 2012	[165]
ISCX 2016	[172]
ADFA	[152]
CAIDA	[173]
ISOT CID	[174]
ISOT HTTP	[175]
Malicious URLs Dataset	[176]
EMBER	[177]
CICDDoS2019	[178]
CICDoS2017	[179]
USTCTFC2016	[180]
GPRS WPA2/WEP	[181]
MTA KDD 19	[182]
LITNET-2020	[183]
CIRA-CIC-DoHBrw-2020	[184]
Bot-IoT	[185]
UCI / Kaggle Datasets such as N-BaIoT	[186,187]

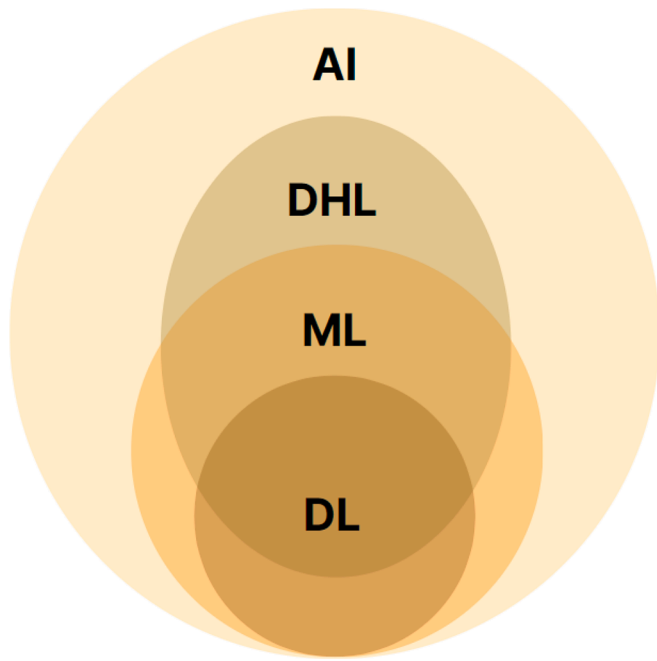


Fig. 8. Relationship between AI, DL, ML, and DHL.

GPRS-WPA2, and found their system to be highly effective, with accuracies ranging from 82.89 % to 99.99 %. Their future work involves exploring a broader range of validation methods given the consistent performance. Alrashdi et al. [148] developed an IoT system for anomaly detection in smart cities using the RF model. They utilized the UNSW-NB15 dataset featuring 49 attributes and nine attack categories, achieving an impressive accuracy rate of 99.34 %. Their future work intends to utilize distributed computing resources to scale the model across fog nodes and improve its performance metrics. The intersection

of IoT and cloud computing creates an environment that amplifies the capabilities of both technologies. IoT devices, which are often resource-constrained, benefit from the scalable and powerful computational resources available in cloud environments. This collaboration allows for the efficient storage and advanced analysis of the massive data volumes generated by IoT devices. Cloud infrastructure enables real-time monitoring and control over these devices, facilitating instant decision-making and automated responses based on data analytics. Moreover, the cloud provides a unified platform that enhances interoperability among a diverse array of IoT devices, making it easier for organizations to manage and extract valuable insights from their connected systems. Given the escalating number of IoT devices, there is an immediate need for effective detection mechanisms to identify and remove compromised devices. As we transition into a hyper-connected era, the security of these IoT devices becomes paramount.

Utilizing real-world cloud data, Alshammari and Aldribi [149] constructed ML algorithms aimed at identifying harmful traffic in cloud computing frameworks. They used the publicly accessible ISOT CID dataset, which consisted of 17,296 training instances and 7,411 for testing. The primary objective was to augment the existing feature set and deploy various models such as DT, RF, SVM, KNN, and NN against the dataset. The complete dataset contained 89,364 entries, 44,569 of which were malicious and 44,795 were benign. Multiple cross-validation folds (5, 10, 15) and different data splits (90–10 %, 80–20 %, 70–30 %) were utilized for evaluation. In all cross-validation folds, DT, RF, and KNN models achieved 100 % accuracy, while during the various data split scenarios, both DT and RF similarly recorded 100 % accuracy. Sethi et al. [150] introduced an IDS that utilizes Double Deep Q-learning (DDQN) to shield cloud networks against cyber threats. They employed both the ISOT CID dataset and the widely-used NSL-KDD dataset. Of the complete 8 TB ISOT CID dataset, only the network traffic segment was considered for the study. Feature selection was done using the chi-square algorithm, choosing 164 features for ISOT CID and 36 for NSL-KDD. In terms of performance, the DDQN-based IDS model yielded an accuracy of 83.40 % with the NSL-KDD dataset and 96.87 % with the

Table 8
The most utilized algorithms in the literature.

Algorithm	Papers where it was used	Threat Domain	Highest Accuracy
SVM	[24,281,282,283,75,78,82,86,88,89,92,98,265,269,149,284,153,285,286,287,288,289,290,291,292,293,177]	Insider Threat, DDoS, Malware, Botnet, Malicious Traffic, Phishing	93.95 %
DT	[77,83,86,88,92], [177,98,149,153,286,289,294,295]	Insider Threat, DDoS, Phishing, Malware, Botnet, Malicious Traffic	100 %
RF	[75,76,77,79,83,88,92,94,177,98,148,268,296,274,149,153,297,298,285,299,300,301,286,302,293,290],	DDoS, Phishing, Malware, Botnet, Malicious Traffic,	100 %
NB	[79,82,86,88,92,98,296,149,153,298,285,299,301,286,287,289,294,303,304]	DDoS, Malware, Botnet, Malicious Traffic, Phishing	90 %
KNN	[98,149,153,298,301,286,305]	Botnet, Malicious Traffic,	100 %
MLP	[67,78,79,80,86,88,177,99,153,299]	DDoS, Malware, Botnet, Phishing, Malicious Traffic	99.60 %
ELM	[177,144,290]	Malicious Traffic	99.5 %
LR	[77,177,268,274,298,285,286,300],	DDoS, Malware, Malicious Traffic,	99.92 %
ANN	[84,101,280,306,286,307]	Phishing, Zero-Day	99.6 %
RNN	[282,66,69,71,81,82,260,154278,308]	Insider Threat, DDoS, Malicious Traffic,	100 %
CNN	[70,71,81,93,143,145,269,278,308,293]	Insider Threat, DDoS, Malicious Traffic, Malware	99 %
DNN	[282,68,147,262,309]	Insider Threat, DDoS, Malicious Traffic, Monitoring Traffic	99.99 %
LSTM	[68,71,73,89,99,262,310]	DDoS, Botnet, Malicious Traffic, Phishing	99 %
AE	[69,74,277]	DDoS	99 %

Table 9

List of all algorithms that were deployed.

ML	DL	DHL
LR	CNN	CNN tried with different GB variants
DT	ALSTM	
RF	LSTM	FCN tried with different GB variants
KNN	FCN	
SGD	DBN	
NB	GRU	
LDA	CNN – LSTM	CNN – LSTM tried with different GB variants
QDA	RNN	
XGB *	PNN	MLP – LSTM and MLP – ALSTM tried with different GB variants
CatBoost *	MLP – LSTM	
AdaBoost *	MLP – ALSTM	
LightGBM	RBF	
SVM	LVQ	
MLP	FCN – LSTM	FCN – LSTM and FCN – ALSTM tried with different GB variants
	FCN – ALSTM	
	RBM	

* A variant of GB algorithms

ISOT CID. Upon adversarial testing, the model recorded accuracies of 79.77 % and 92.17 % for NSL-KDD and ISOT CID, respectively. Xie et al. [151] implemented an SVM based short sequence algorithms. They utilized the ADFA dataset [152], which is comprised of benign and anomalous traffic stemming from six different types of cyber-attacks. For this study, duplicate records were omitted, improving the distinction between normal and abnormal traffic. Different k-values (3, 5, 8, 10) were tested, with $k = 5$ producing the best outcomes. The model displayed 70 % accuracy. Vanhoenshoven et al. [153] explored various techniques for detecting malicious URLs. They used a dataset known as the Malicious URLs Dataset, which includes 121 sets of data gathered over a period of 121 days. The dataset features a staggering 2.3 million URLs and 3.2 million attributes. The researchers classified these URLs into three distinct groups based on specific characteristics. Different models, including MLP, DT, RF, and KNN, were assessed using various performance metrics such as accuracy, precision, and recall. According to the study, all applied methodologies showed high accuracy, with the RF model being particularly effective with an approximate accuracy of 97 %. Yuan et al. [154] proposed a parallel neural joint model strategy to identify harmful URLs more efficiently. The model integrated semantic and text attributes through a parallel joint NN that included both a Capsule Network (CapsNet) and an Independent Recurrent Neural Network (IndRNN). Data for malicious URLs were sourced from Phish-Tank, an anti-phishing platform, and a blacklist of dangerous websites. They employed a 5-fold cross-validation method and utilized standardized performance metrics to assess the model. The study revealed that the model performed optimally when the feature dimension was 185 and the IndRNN layers were set at 2. The model's accuracy and recall were both remarkably high, reaching 99.78 % and 99.98 %, respectively, surpassing the performance of traditional models. In terms of scalability, both the cloud and IoT effortlessly adapt to increased data and computational demands. Additionally, cloud platforms offer robust security features, including encryption and authentication, to safeguard the data transmitted from IoT devices against unauthorized access and cyber threats. The cost-effectiveness of leveraging cloud resources also eliminates the need for substantial investments in dedicated data centers, making IoT-cloud integration particularly advantageous for small to medium-sized enterprises.

Choosing the appropriate dataset and features or ML and DL algorithms for identifying various attack types remains a complex decision for practitioners. Research often involves comparing algorithms to pinpoint the best approach for building an intelligent model. As there's no one-size-fits-all algorithm, the need for domain-specific algorithm is evident. The ultimate aim is to translate these models into real-time

systems for practical applications. Real-time ML comprises two facets: online predictions and online learning. Both are essential for a system to adapt dynamically to new data. Therefore, converting these intelligent models into real-time NIDS presents a crucial direction for future research.

6. The available datasets

IoT and CPS are two interrelated concepts that often intersect in the realm of interconnected and smart systems. IoT refers to the network of physical devices found in household appliances, vehicles, and industrial machines, and embedded with sensors, software, and other technologies to collect and exchange data over the internet. CPS, on the other hand, is a broader term that encompasses the integration of computer-based algorithms with physical processes. These systems are designed to monitor and control physical elements through computing and networking capabilities [155]. These systems find applications in a multitude of sectors like industrial automation, energy management, autonomous vehicles, public transport, and healthcare. One characteristic that sets CPSs apart is their mission-critical nature, necessitating a high degree of fault tolerance. Over recent years, the development trend in CPS has moved away from proprietary platforms toward standardized, open-source hardware, software, and communication protocols. This shift is spurred by advantages such as quicker development timelines, cost-efficiency in both development and upkeep, and enhanced system compatibility. However, this open-source paradigm also enlarges the potential for cybersecurity vulnerabilities in systems once considered intrinsically secure. Substantial research efforts are currently aimed at fortifying this new breed of open-source CPS against security threats. As AI, specifically DL, has burgeoned, its role in safeguarding CPS has gained traction in both academic and industrial circles [156]. However, the effectiveness of modern DL algorithms is contingent upon the availability of extensive training data. Herein lies a challenge: obtaining large-scale, representative CPS datasets is difficult. Firstly, traditional IT security datasets don't accurately represent the dynamics specific to CPS. Secondly, the large datasets that do exist within actual CPS operations, such as in manufacturing or healthcare, are often withheld due to security concerns. To address this, scholars often resort to creating their own data by constructing either scaled-down physical labs or virtual testing environments [157]. Even though some researchers make their datasets public, these datasets frequently lack essential attributes required for comprehensive academic study. Thus, the lack of standard guidelines on dataset characteristics and formats may impede academic progress, confining research to organizations with access to substantial, confidential datasets. Both IoT and CPS aim to enhance the interaction between the physical and digital worlds, but they serve different ends in this objective. IoT focuses primarily on the data collection and communication aspects, essentially providing the "sensory organs" that enable systems to perceive the environment. CPS, in contrast, aims to create a responsive system that not only collects data but also processes it to control physical elements, serving as both the "brain" and the "muscles" in the system. CPS often relies on the data and communication infrastructure provided by IoT to make real-time decisions and control physical systems. In essence, IoT can be viewed as a subset or a component within a CPS. The CPS leverages IoT data to make intelligent decisions that are critical for mission-centric operations like manufacturing process controls, autonomous driving, or medical systems. Therefore, IoT and CPS are often closely integrated in modern smart systems, with IoT providing the data and CPS offering the control mechanisms to act on that data. Table 6 summarizes some of the qualities [158] that are sought after in IoT Networks Datasets.

Creating datasets for IoT devices is an intricate and time-consuming task [159]. Even though there have been strides in the development of automated testing tools that can execute or simulate many attacker actions, the majority of the process remains reliant on human intervention. This human dependency leads to delayed releases of dataset iterations.

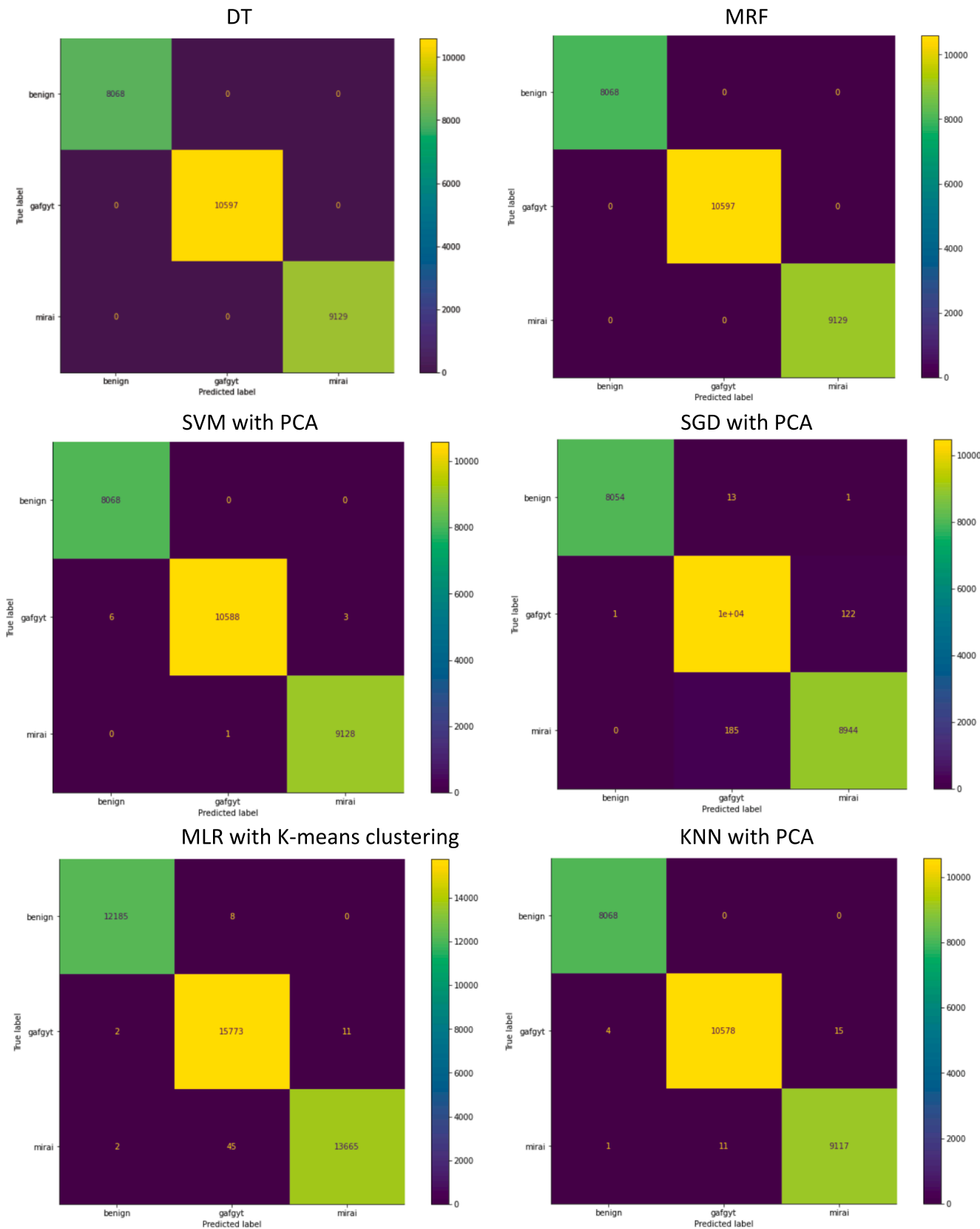


Fig. 9. Confusion matrix values for the DANMINI video Door Phone Doorbell Hands Free Wireless Intercom for DT, RF, SVM, SGD, MLR, and KNN models.

Table 10

Average accuracy, precision, and recall values for DANMINI doorbell for ML models.

Model	Accuracy	Precision	Recall
RF	90.86133697920414 %	92.59459525341209 %	90.86133697920414 %
RF with PCA	99.96042311290206 %	99.96043377376141 %	99.96042311290206 %
DT	100.00 %	100.00 %	100.00 %
DT with PCA	99.70497229617903 %	99.7050456821072 %	99.70497229617903 %
SVM	99.96402101172915 %	99.96403669985497 %	99.96402101172915 %
SVM with PCA	99.96402101172915 %	99.96403669985497 %	99.96402101172915 %
SGD	98.84147657767863 %	98.84345112553179 %	98.84147657767863 %
SGD with PCA	91.70684320356911 %	91.72961993760130 %	91.70684320356911 %
LR	99.79851766568324 %	99.79909184732848 %	99.79851766568324 %
LR with K-means	99.83689525317214 %	99.83714395981383 %	99.83689525317214 %
LR with PCA	99.79851766568324 %	99.79909184732848 %	99.79851766568324 %
KNN	99.88846513636036 %	99.88846410721016 %	99.88846513636036 %
KNN with PCA	99.88846513636036 %	99.88846410721016 %	99.88846513636036 %
NB	70.17341872346550 %	69.96064372464190 %	70.17341872346550 %
XGB	100.00 %	100.00 %	100.00 %
XGB with PCA	99.95682521407498 %	99.95682521407498 %	99.95682521407498 %
AdaBoost	99.98920630351874 %	99.98920935833850 %	99.98920630351874 %
AdaBoost with PCA	99.20126646038714 %	99.20129736087069 %	99.20126646038714 %
LightGBM	99.99640210117291 %	99.99640249524726 %	99.99640210117291 %
LightGBM with PCA	99.96761891055623 %	99.96761868866684 %	99.96761891055623 %
LDA	91.20673526660430 %	91.32998392363270 %	91.20673526660430 %
LDA with PCA	91.20673526660430 %	91.32998392363270 %	91.20673526660430 %
QDA	88.36439519320717 %	90.86777804855624 %	88.36439519320717 %
QDA with PCA	88.36439519320717 %	90.86777804855624 %	88.36439519320717 %

Achieving complete automation, the introduction of disturbances, attack formulation and execution, as well as automated data collection and tagging, remains a long-term objective [160].

Several datasets are commonly employed in the research literature for evaluating network detection and classification models. NSL-KDD stands out as the most popular due to its manageable size and public availability, featuring 41 variables and being an optimized version of the original KDD dataset. Following NSL-KDD are KDD-99, an upgraded version of DARPA98, and CICIDS2017, released by the Canadian Institute for Cybersecurity, which includes both standard and newer attack types. The UNSW-NB15 dataset, generated using the IXIA tool, encompasses nine categories of attacks and is another frequently used resource. Some researchers have opted to develop their own datasets, like CTU-13 from the Czech Technical University, which features real botnet and normal traffic across thirteen scenarios, including DoS attacks. The SNMP-MIB dataset has nearly 5,000 entries with 34 variables, capturing six types of DoS and web brute force attacks. The Kyoto 2006 + dataset was created using real-world traffic data and comprises 24 features. Table 7 shows a summary of the dataset that were utilized by various scholars in this field. Most of these datasets are related to network security monitoring and data collection.

In IDS, data can be sourced in two primary ways: either by leveraging pre-existing datasets or by generating a unique dataset. Various existing datasets are routinely employed in the field of Anomaly Detection. Among these, KDD-99 [188] has an enduring presence despite its known imbalance issues, largely due to the absence of suitable alternatives. NSL-KDD [189] was developed as an improvement over KDD-99, addressing some of its limitations. However, it remains a standard for anomaly detection even if the attack vectors it contains are somewhat dated. In contrast, UNSW-NB15 [164,190], mitigates the data imbalance found in KDD-99 and NSL-KDD, featuring 49 attributes and 10 categories of attacks, compared to KDD-99's 41 features and 5 classes. For studying Botnets, CTU13 [191] is currently favored, featuring 13 distinct scenarios that involve different kinds of Botnet samples. Each scenario involves running a unique malware piece that operates through various protocols. Additionally, the ISOT dataset [192] is popular, especially for databases focused on IoT Botnet attacks. Multiple tools and methods are available for those looking to generate their datasets. Wireshark, for instance, is effective for capturing and displaying network traffic. Its utility is emphasized when dealing with DoS attacks, as the patterns are rather predictable. Spleen is another software

platform akin to DARPA [193], which also allows for the addition of extra features for enhanced utility. CICFlowmeter[194] is a Java-based tool that extracts 80 distinct network attributes from raw traffic, generating either PCAP or CSV files for further analysis [194]. A limitation of this dataset is the small number of systems used, necessitating updates with newer attack vectors and additional Botnet data. Meidan et al. [195] conducted experiments involving infecting commercial IoT devices in a controlled lab setting using Mirai and Bashlite Botnets. Network traffic was captured using port mirroring on the central switch and recorded with Wireshark. In a similar vein, Djanie et al. [195] utilized eight DoS attack tools on a virtual machine running Ubuntu 19. When choosing a dataset for scholarly work, the number and volume of attributes are a critical determinant. Dataset quality significantly influences the performance of NIDS. Large, well-distributed datasets are recommended for better accuracy. However, the limited availability of such datasets poses a challenge. KDD-99 and its successor NSL-KDD are commonly used, but other options like ADFA, ISOT HTTP, and IoT20 offer further possibilities.

It is worth mentioning that some of the datasets listed in Table 7 do not contain IoT traffic while others might be outdated. However, we decided to include them because they all contain traffic data that were used to advance the knowledge related to detecting cybersecurity threats.

7. Methodology

AI is an expansive field dedicated to emulating human cognitive abilities in machines [196,197,198,199]. Originally termed by John McCarthy in 1955 [200,201]. AI aims to design machines capable of independent reasoning, data analysis, and decision-making [202,203,204]. It serves as an interdisciplinary field focusing on the science and engineering necessary for machines to perform tasks generally considered to require human cognition [205]. Beyond its technical scope, AI embodies the broader notion that machines can successfully engage in activities usually dependent on human intellect [206]. Today, AI's presence is increasingly felt in various consumer technologies such as Google Home, Siri, Alexa, and Open AI applications [207,208]. It is the driving force behind predictive algorithms in streaming services such as Netflix and Amazon Prime. Moreover, AI algorithms play a pivotal role in financial hedge funds, enabling micro-trades that amass significant revenue [209]. ML, a significant subset of

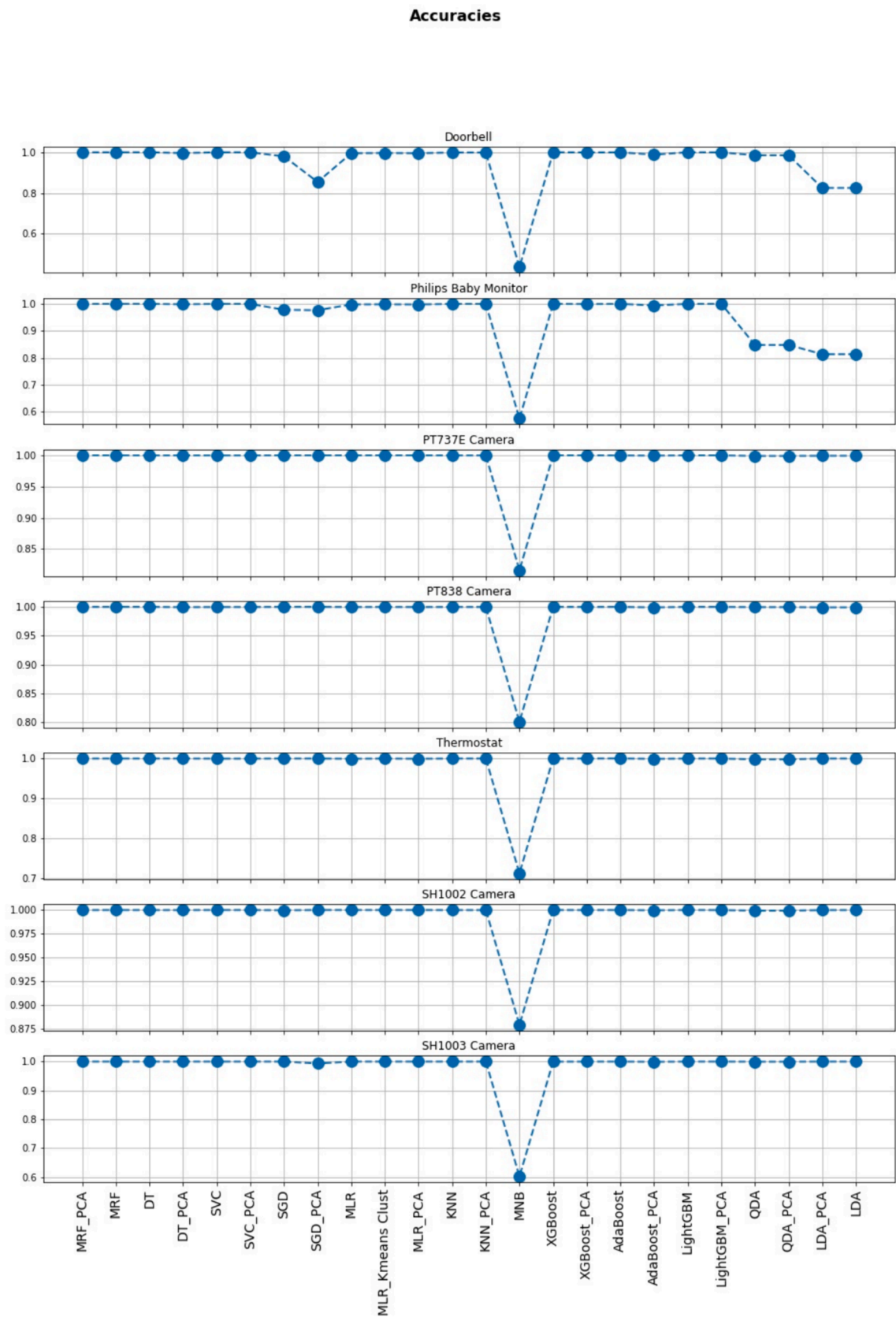


Fig. 10. ML models' accuracies when detecting Mirai botnet attacks.

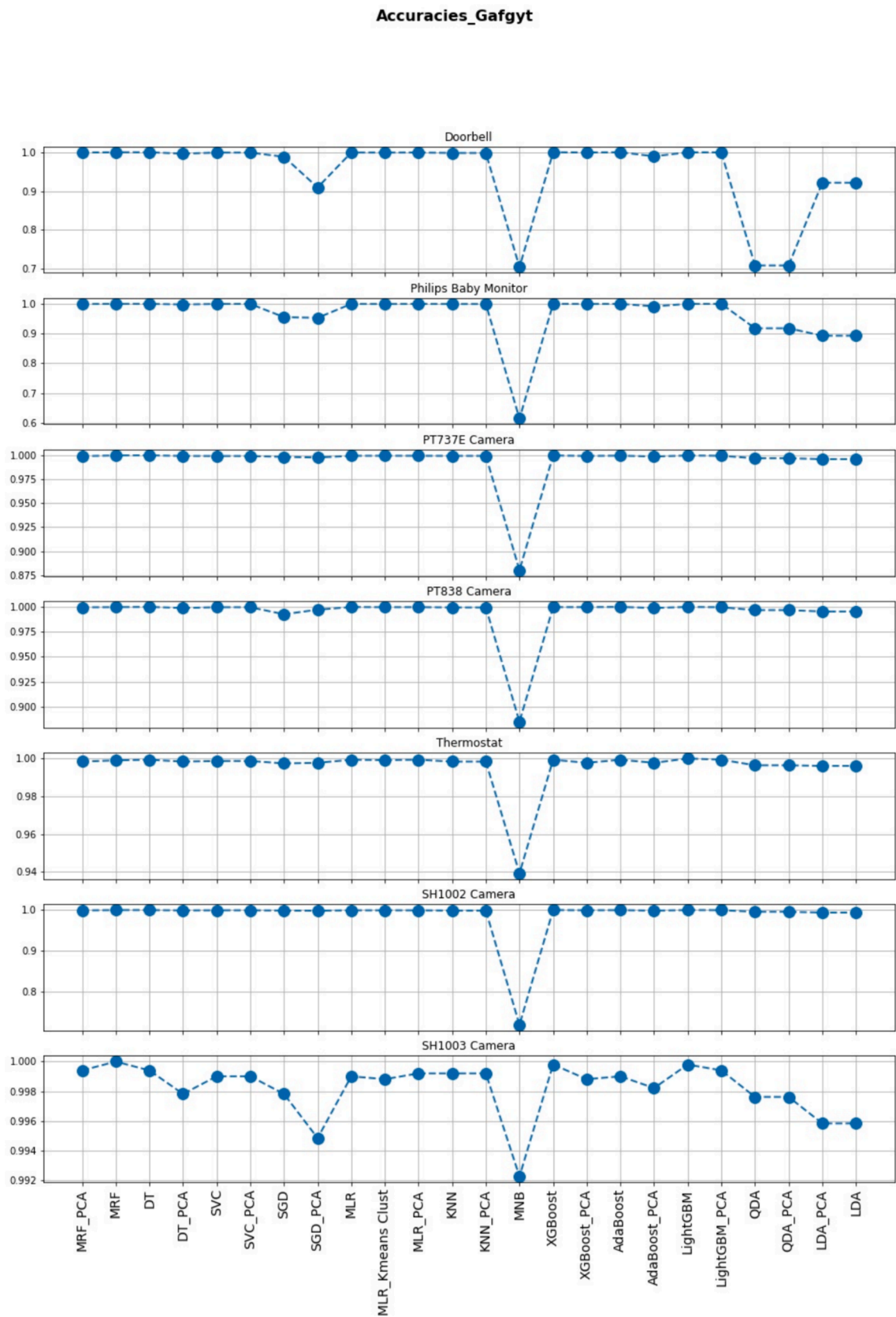


Fig. 11. ML models' accuracies when detecting Gafgyt botnet attacks.

Table 11

Average detection accuracy for Mirai and Gafgyt botnet attacks in each device.

Botnet Attack	Doorbell	Philips Baby Monitor	PT737E Security Camera	PT838 Security Camera	Thermostat	Home 1002 Security Camera	Home 1003 Security Camera
Mirai	95.24 %	95.15 %	99.22 %	99.15 %	98.75 %	99.49 %	98.28 %
Gafgyt	95.17 %	96.35 %	99.39 %	99.37 %	99.58 %	98.69 %	99.82 %

AI, focuses on the development of algorithms that improve performance through ongoing data analysis [197]. It empowers machines to learn autonomously by exposing them to large datasets. Unlike ML, which concentrates on specific tasks, AI remains a dynamic and evolving field, continually adapting as technology progresses [210]. It is an umbrella term that includes various subfields such as ML and DL [211,212]. AI is particularly transformative in automation technologies [213]. ML excels in sifting through extensive data, identifying recurring patterns and intricate details [214]. It employs data to build predictive models automatically, aiming to make decisions without human oversight [215]. Initially, the system uses probability-based decision-making, refined through feedback loops to improve the accuracy of its predictions. With each iterative cycle, the machine becomes more competent and intelligent [216]. Deep Hybrid Learning (DHL) is an advanced technique that merges DL methodologies with various other ML methods within a singular model [217]. DHL employs multiple-layer DL networks to model intricate patterns in data and integrates different types of ML algorithms. This can encompass a mix of supervised and unsupervised learning techniques, different NN architectures, or an array of statistical and machine learning strategies. Fig. 8 shows the relationship between AI, DL, ML, and DHL [218,219].

7.1. Background

Data preprocessing is an integral part of ML workflows, consisting of several operations such as filling in missing values, normalizing datasets, and filtering out irrelevant features or outliers [220,221]. In this segment, an overview of scholarly works employing intelligent algorithms to identify various types of cyberattacks is provided [222]. A diverse range of ML methodologies have been utilized in these studies, encompassing classification, regression, and clustering techniques [223,224]. Among these, LR and DT are notable. RF, a composite of multiple DT, has been used by some researchers to graphically illustrate the decision-making flow using tree-like structures. SVM has garnered considerable attention in the realm of classification, particularly for its prowess in clearly segregating data points by constructing a hyperplane within an n-dimensional feature space. NB is another classifier that has seen widespread application; it's a supervised learning algorithm based

Table 12

Average accuracy, precision, and recall values for DANMINI doorbell for all DL models.

Model	Accuracy	Precision	Recall
RNN	90.39 %	90.40 %	90.38 %
LSTM	94.76 %	94.77 %	94.73 %
ALSTM	88.34 %	88.34 %	88.33 %
MLP	99.31 %	99.31 %	99.31 %
MLP-LSTM	95.28 %	95.29 %	95.28 %
MLP-ALSTM	84.75 %	84.75 %	84.75 %
CNN-LSTM	95.56 %	95.57 %	95.55 %
CNN-ALSTM	95.06 %	95.06 %	95.06 %
2-CNN	99.95 %	99.95 %	99.95 %
2CNN-LSTM	88.98 %	88.99 %	88.98 %
2CNN-ALSTM	94.74 %	94.74 %	94.73 %
LSTM-FCN	90.92 %	90.92 %	90.92 %
ALSTM-FCN	90.93 %	90.92 %	90.92 %
DBN	48.90 %	48.91 %	48.89 %
PNN	92.73 %	92.74 %	92.75 %
RBFN	93.56 %	93.52 %	93.52 %
LVQ	76.57 %	76.55 %	76.59 %

on the principles of Bayes' theorem. Lastly, KNN algorithm has been employed for classification tasks, and K-means has been used in unsupervised learning contexts for clustering [225]. Feature extraction is central to ML models, employing various optimization techniques such as PCA, GA, and Boosting Algorithms. Due to their computational efficiency, ML methods are often preferred for feature engineering. When focusing on botnet analysis, it is typically divided into two primary categories: Flow-based traffic analysis and Graph-based traffic analysis [226]. The distinction between these two lies mainly in the types of features selected. Flow-based analysis usually opts for statistical features, while Graph-based methods go for graph-specific features. For real-time analysis, the same features are often examined across different time intervals [195]. These features are especially effective in identifying anomalous behavior, such as those resulting from a spoofed IP address [227]. Chowdhury et al. have effectively detected botnets using Graph-based features by eliminating inactive nodes, thereby improving detection efficiency [131]. In the realm of IDS, algorithms for classification and clustering such as SVM are extensively employed [228,229], DT [230,231], KNN [232] have produced noteworthy results when analyzing network traffic. Incremental Learning (IL) techniques have recently garnered attention for providing real-time results, although traditionally they have been more common in fields like image recognition and target identification [233,234]. SVM has been notably for intrusion detection [235]. Additionally, Yi et al. [236] designed an enhanced Incremental SVM algorithm that solves oscillation issues in the learning process. This approach was further utilized by Zhuang et al. for malware detection [237,238]. Despite the benefits of SVM in real-time scenarios, its applicability can be limited by the small size of the dataset, affecting its performance. Dromard et al. introduced a real-time, unsupervised network anomaly detector using Incremental Grid Clustering (IGC) [239]. Their study demonstrated the scalability of their approach but noted potential for speed improvements. Issues with false positives in real-time anomaly detection were observed by Landauer et al., using an algorithm that self-learns and creates clusters over time [240]. Yang et al. used damped incremental statistics to reduce memory usage [241]. GA and swarm intelligence techniques have also started to gain traction for optimization tasks. For example, Shorman et al. achieved significant outcomes by employing Grey Wolf Optimization (GWO) for hyperparameter tuning in one-class SVM [242]. Various DL methods have been utilized for constructing cyberattack detection models. A predominant focus has been on ANN, which excel in handling nonlinear dependencies through multiple interconnected layers. RNN, a specific ANN variant, incorporate memory functions to preserve previous states and therefore are effective for sequential data. CNN serves as another key DL technique, tailored to simulate human visual recognition systems. DNN is a supervised learning-oriented ANNs adept at mapping inputs to outputs through intricate mathematical transformations. LSTM networks, specialized RNNs designed to model sequential data more effectively, and MLP, an ANN configuration with multiple layers of directed graphs, are also widely employed. GRU, a more efficient LSTM variant in terms of memory utilization and computation time, have garnered attention as well [243]. Feature engineering is particularly enhanced by DL techniques. Generative Adversarial Networks (GAN) are popular for synthetic data generation and understanding minority classes better, as seen in Ferdowsi et al. [244]. Studies by Lee et al. [245], Han et al. [246], and Yang et al. [247] showcase innovative approaches to feature engineering, including Attention GRU models and Supervised adversarial Variational autoencoders. Studies have also

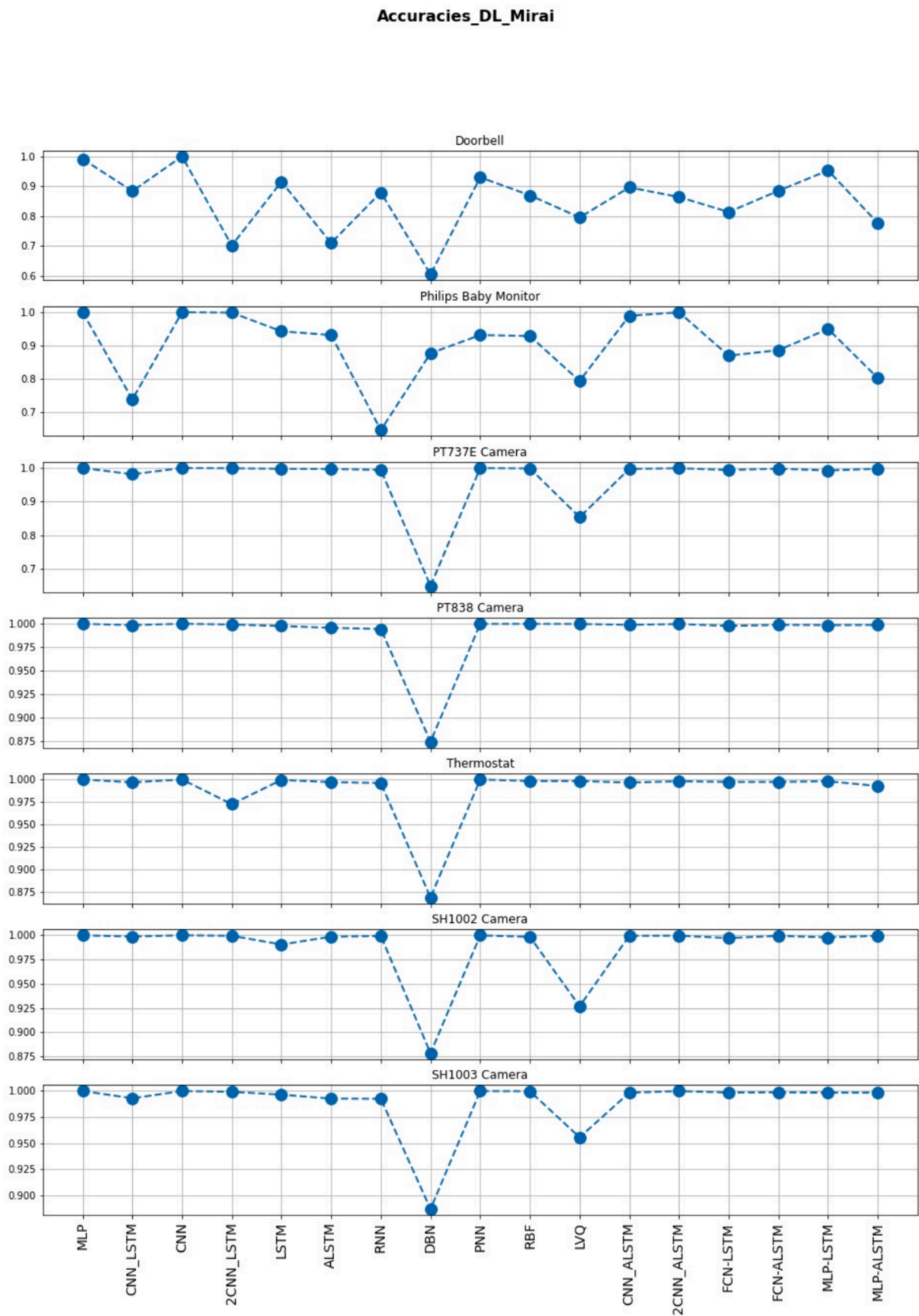


Fig. 12. DL models accuracies when detecting Mirai botnet attacks.

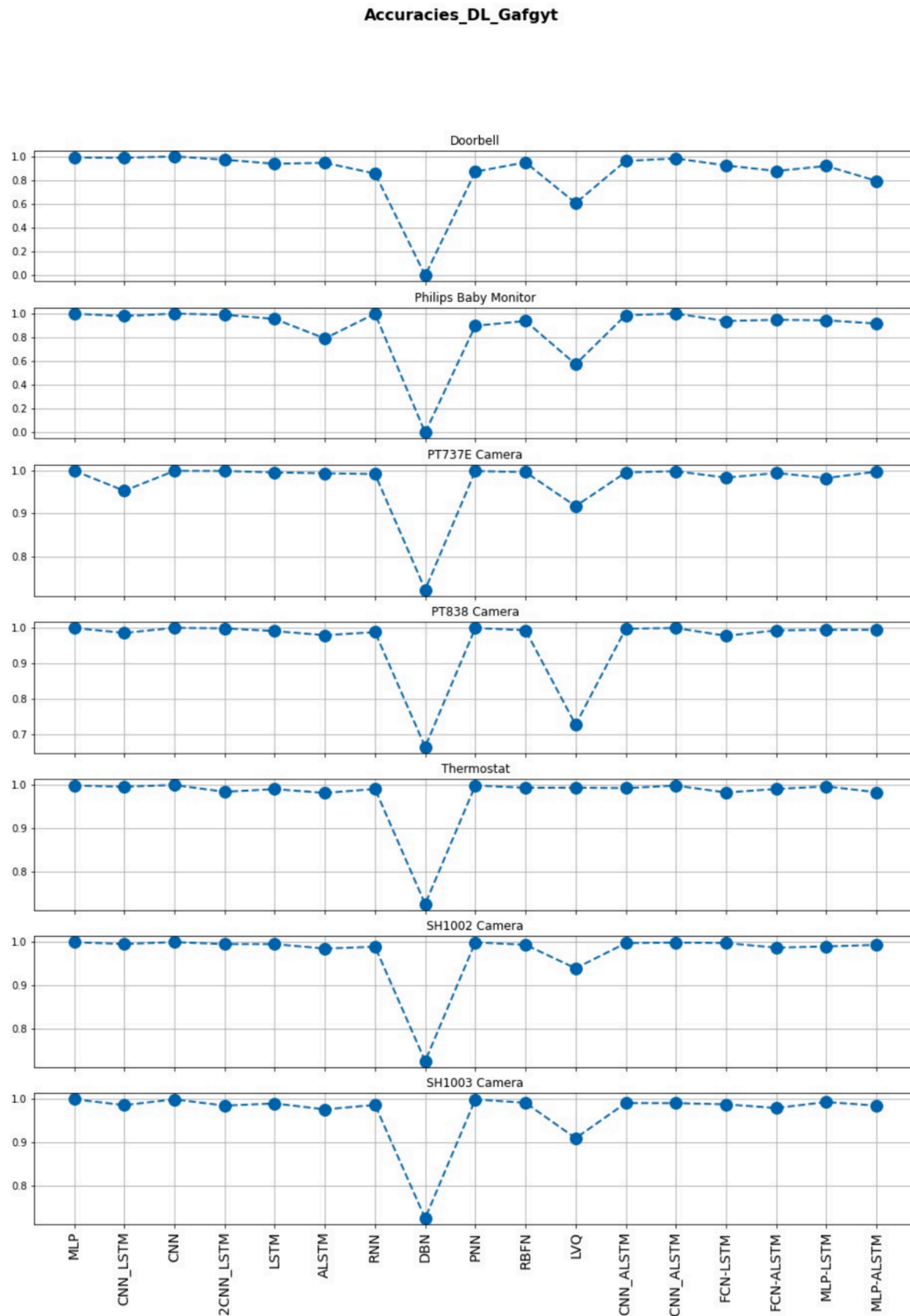


Fig. 13. DL models' accuracies when detecting Gafgyt botnet attacks.

addressed specific challenges. For instance, Jiang et al. [248] focus on data imbalance, employing techniques like One-Side Selection (OSS) and Synthetic Minority Over-Sampling (SMOTE) for more equitable representation. Others, like Martin et al. [249] employ Deep Reinforcement Learning (DRL) algorithms for fast responses in online learning contexts. Kumar and Bhama [250] apply Sparse Autoencoders (SAE) and outlier detection to preemptively identify IoT Botnets, whereas Otoum et al. [251] utilize Restricted Boltzmann Machines

(RBM) for similar aims. Studies by Wei et al. [252], Xu et al. [253], Zhong et al. [254], and Muhuri et al. [255] compare various DL techniques and traditional ML methods in feature engineering and detection tasks. Ensemble methods, such as those employed by Chen et al. [256] and Liu and Yin [257], show promise in addressing complicated attack vectors like link-flooding DDoS attacks. Additional work by Zavrak and Iskefiyeli [258] indicates that Variational Autoencoders (VAE) yield similar Receiver Operating Characteristic (ROC) curves for attacks with

Table 13

Average detection accuracy for Mirai and Gafgyt botnet attacks in each device.

Botnet Attack	Doorbell	Philips Baby Monitor	PT737E Security Camera	PT838 Security Camera	Thermostat	Home 1002 Security Camera	Home 1003 Security Camera
Mirai	85.13 %	89.87 %	96.73 %	99.12 %	98.84 %	98.72 %	98.85 %
Gafgyt	85.70 %	87.35 %	97.18 %	95.75 %	97.57 %	97.50 %	96.91 %

Table 14

The average detection accuracy of ML and DL models in all IoT devices (Continued).

Model	Doorbell	Thermostat	Monitor	PT737E	PT838	SH1002	SH1003
LR	64.04 %	61.27 %	55.59 %	52.64 %	51.30 %	59.47 %	60.51 %
RF	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %
DT	100.00 %	99.99 %	100.00 %	99.99 %	100.00 %	99.99 %	99.99 %
SGD	67.85 %	68.98 %	58.51 %	54.98 %	38.32 %	16.93 %	68.16 %
kNN	99.99 %	99.99 %	99.99 %	99.97 %	99.98 %	99.98 %	99.98 %
NB	65.91 %	63.28 %	57.34 %	54.82 %	53.48 %	61.46 %	62.52 %
XGB	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %
LDA	99.74 %	98.03 %	99.49 %	99.47 %	99.70 %	97.97 %	98.03 %
QDA	80.02 %	74.90 %	81.08 %	72.45 %	74.06 %	76.64 %	75.20 %
AdaBoost	94.11 %	99.98 %	99.97 %	99.77 %	97.27 %	99.94 %	99.99 %
LightGBM	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %
MLP	99.99 %	99.98 %	99.99 %	99.97 %	99.97 %	99.98 %	99.98 %
CatBoost	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %
SVM	99.96 %	99.91 %	99.99 %	99.94 %	99.94 %	99.97 %	99.92 %
SVC_CV_PCA	99.96 %	99.91 %	99.97 %	99.95 %	99.94 %	99.97 %	99.92 %
LR_CV_PCA	99.92 %	99.72 %	99.90 %	99.75 %	99.87 %	99.71 %	99.72 %
RF_CV_PCA	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %	99.99 %	99.99 %
DT_CV_PCA	99.99 %	99.99 %	99.98 %	99.97 %	99.98 %	99.99 %	99.98 %
SGD_CV_PCA	99.90 %	99.69 %	99.69 %	99.79 %	99.84 %	99.67 %	99.64 %
KNN_CV_PCA	99.98 %	99.98 %	99.97 %	99.97 %	99.97 %	99.97 %	99.97 %
NB_CV_PCA	69.98 %	64.39 %	59.41 %	57.62 %	59.71 %	63.83 %	63.32 %
XGB_CV_PCA	100.00 %	99.99 %	99.99 %	100.00 %	100.00 %	99.99 %	99.99 %
LDA_CV_PCA	97.43 %	95.52 %	97.71 %	98.23 %	97.98 %	95.59 %	95.08 %
QDA_CV_PCA	79.61 %	74.94 %	81.68 %	72.79 %	75.63 %	76.60 %	75.01 %
AdaBoost_CV_PCA	94.44 %	89.01 %	91.27 %	89.72 %	93.84 %	91.21 %	88.60 %
LightGBM_CV_PCA	100.00 %	99.80 %	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %
MLP_CV_PCA	99.99 %	99.99 %	99.99 %	99.99 %	99.98 %	99.98 %	99.98 %
CatBoost_CV_PCA	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %	100.00 %	99.99 %
PNN	92.74 %	99.91 %	94.09 %	99.92 %	99.92 %	99.92 %	99.94 %
2CNN_LSTM	99.98 %	99.98 %	99.97 %	99.96 %	99.97 %	99.98 %	99.97 %
ALSTM	99.99 %	99.99 %	99.97 %	99.97 %	99.97 %	99.97 %	99.96 %
CNN	67.14 %	67.14 %	57.98 %	55.84 %	54.75 %	61.58 %	62.80 %
CNN_LSTM	99.99 %	99.99 %	99.99 %	99.97 %	99.98 %	99.98 %	99.98 %
DBN	72.41 %	72.41 %	63.80 %	72.70 %	69.27 %	74.55 %	77.85 %
FCN_LSTM	66.98 %	66.98 %	57.94 %	54.82 %	53.47 %	61.64 %	62.81 %
FCN_ALSTM	64.01 %	64.01 %	55.53 %	52.54 %	51.28 %	59.41 %	60.70 %
FCN_2ALSTM	64.01 %	64.01 %	55.53 %	61.26 %	51.28 %	59.41 %	60.70 %
GRU	99.99 %	99.99 %	99.99 %	99.98 %	99.98 %	99.98 %	99.98 %
LSTM	99.99 %	99.99 %	99.98 %	99.97 %	99.97 %	99.97 %	99.95 %
MLP_LSTM	65.82 %	65.82 %	57.12 %	54.81 %	53.46 %	59.41 %	62.79 %
MLP_ALSTM	64.01 %	64.01 %	55.53 %	52.54 %	51.28 %	59.41 %	60.70 %
RBF	81.89 %	81.89 %	85.00 %	85.43 %	78.74 %	78.79 %	82.86 %
LVQ	93.00 %	89.00 %	86.00 %	86.00 %	86.00 %	80.00 %	81.00 %
RBM	66.02 %	63.26 %	57.31 %	54.81 %	53.58 %	61.38 %	62.49 %
RNN	94.56 %	94.56 %	71.87 %	66.44 %	91.19 %	59.41 %	67.82 %
FCN	99.79 %	99.44 %	99.83 %	99.70 %	99.68 %	99.16 %	98.71 %
CNN_XGB	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %	100.00 %	99.99 %
CNN_AdaBoost	98.78 %	99.92 %	99.95 %	99.38 %	99.39 %	99.59 %	93.31 %
CNN_lightGBM	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %	100.00 %	99.99 %
CNN_CatBoost	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %	100.00 %	99.99 %
FCN_XGB	100.00 %	100.00 %	100.00 %	99.99 %	99.99 %	100.00 %	100.00 %
FCN_AdaBoost	82.24 %	98.32 %	99.46 %	99.00 %	99.81 %	88.56 %	92.56 %
FCN_LightGBM	100.00 %	99.99 %	100.00 %	99.99 %	99.99 %	100.00 %	100.00 %
FCN_CatBoost	100.00 %	100.00 %	100.00 %	99.99 %	99.99 %	100.00 %	100.00 %
FCN_LSTM_XGB	100.00 %	99.99 %	99.99 %	99.99 %	99.98 %	99.98 %	99.98 %
FCN_LSTM_CatBoost	100.00 %	99.76 %	99.93 %	99.95 %	99.92 %	99.92 %	98.95 %
FCN_LSTM_AdaBoost	99.75 %	98.43 %	84.10 %	92.47 %	88.13 %	94.57 %	97.75 %
FCN_LSTM_lightGBM	95.14 %	99.99 %	99.99 %	99.99 %	99.99 %	99.98 %	99.99 %
CNN_LSTM_XGB	99.99 %	99.98 %	99.99 %	99.98 %	99.98 %	99.99 %	99.97 %
CNN_LSTM_AdaBoost	99.99 %	99.66 %	99.97 %	99.96 %	99.91 %	99.87 %	99.82 %
CNN_LSTM_LightGBM	100.00 %	99.99 %	99.99 %	99.98 %	99.98 %	98.46 %	99.98 %
CNN_LSTM_CatBoost	100.00 %	99.98 %	99.99 %	99.99 %	99.98 %	99.98 %	99.98 %
CNN_ALSTM_XGB	100.00 %	99.99 %	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %
CNN_ALSTM_AdaBoost	99.99 %	99.98 %	99.98 %	99.98 %	97.25 %	99.96 %	99.96 %
CNN_ALSTM_LightGBM	100.00 %	99.99 %	100.00 %	100.00 %	99.99 %	99.99 %	99.99 %
CNN_ALSTM_CatBoost	100.00 %	99.99 %	100.00 %	99.99 %	99.99 %	99.99 %	99.99 %

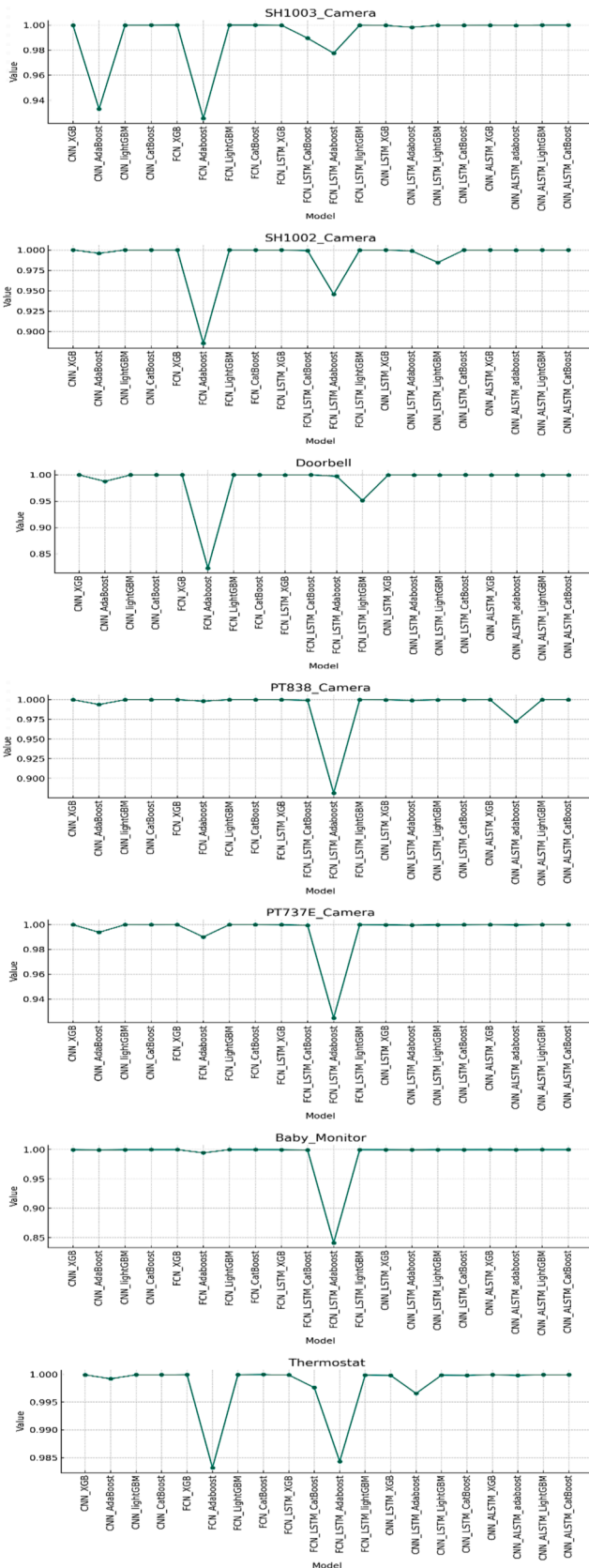


Fig. 14. DHL models accuracies when detecting botnet attacks.

comparable behaviors. Hara and Shiomoto [259] employ Adversarial Autoencoders (AAE) in a semi-supervised learning context, achieving comparable results with significantly fewer labeled data but at the cost

Table 15

Limitations and solutions in detecting security threats in IoT domain.

Limitations	Description	Solution
Robustness	Robustness in the context of ML refers to the ability of a model to yield consistent results across both training and testing datasets. For practical applications in the real world, it's crucial to have a ML model that exhibits this robustness [322]. DHL models are often categorized as robust algorithms [323]. While these models perform well theoretically, not all are widely adopted in enterprise environments, often due to concerns about their robustness in diverse and dynamic real-world scenarios.	By allowing the model to continually adapt and learn, thereby enhancing its robustness. Incremental learning emerges as a viable approach to attain genuine robustness, particularly in dynamic real-world settings where data continuously evolves.
Generalizability	Generalizability in modeling is determined by evaluating a model's performance in unfamiliar testing scenarios, aiming for a model that embodies generalizability is essential for its sustainability [324].	Transfer learning is a technique in ML where a model developed for one task is reused as the starting point for a model on a second task. It's an effective strategy particularly when dealing with a limited amount of data for the new task. In addition, transfer learning can be used to reduce the long training times that ML models need to perform effectively.
Real-time analysis	Most of the current research in NIDS was conducted in an offline settings [325].	Integrating the advantages of both incremental learning and transfer learning could be useful in dynamic real-world settings. Allowing the ML models to continually adapt and learn while leveraging knowledge (features, weights, etc.) from a previously trained model.
Physical limitations of IoT devices	IoT devices often face limitations regarding power, budget, and size. Due to these constraints, fully leveraging the capabilities of ML techniques becomes impractical.	Attaching the edge computing technology to IoT devices to enable it to analyze traffic data at the periphery of the network, close to the source of data generation (i.e., IoT devices), rather than relying on a central data-processing warehouse. This approach reduces latency, decreases bandwidth usage, and allows for quicker, more efficient data processing and decision-making directly at the device level. It enhances the performance and responsiveness of IoT applications by bringing computation and data storage closer to the devices where it's being gathered.

of increased training time. To safeguard both institutions and individuals from cyber threats, it is essential to scrutinize and categorize network data initially, enabling the identification of unusual and malevolent intrusions. Given the paramount significance of classifying malevolent data, numerous scholars have endeavored to enhance classification methodologies harnessing the capabilities of artificial

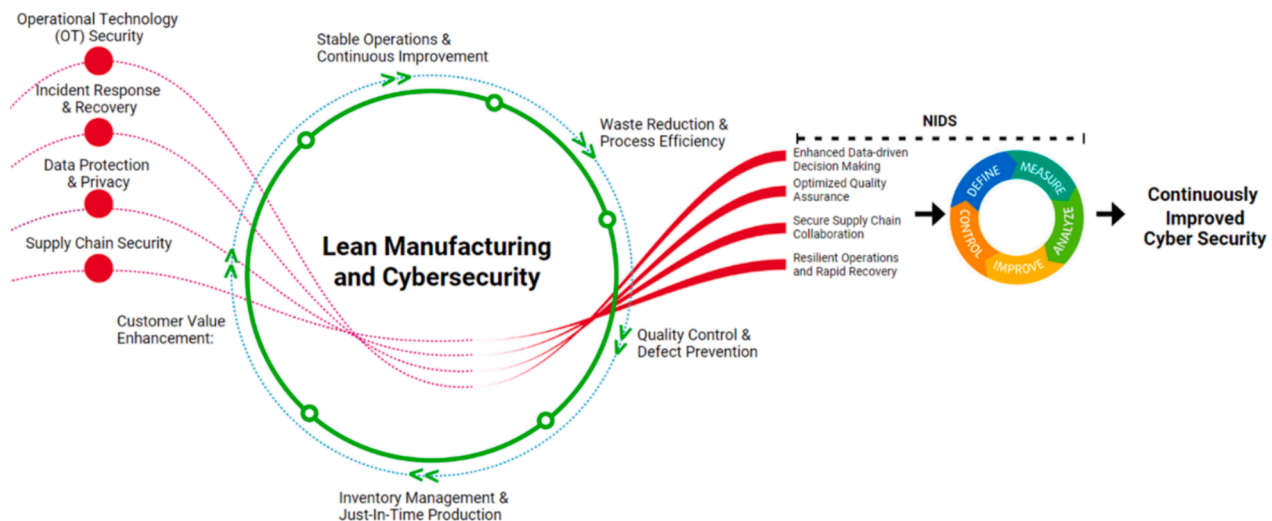


Fig. 15. Cyber resilience and efficient manufacturing.

intelligence. Several investigations have concentrated on identifying irregular and deviant network traffic.

Yang et al. [260] devised a model for uncovering latent aberrant network behavior to identify cyberattacks, employing deep learning techniques. The dataset employed in their investigation comprised information derived from NetFlow campus records, which encompassed data collected by campus routers. In the initial data preparation phase, the authors converted the data into a uniform structure before implementing the RNN algorithm. The resulting model achieved an impressive accuracy rating of 98 %, the authors plan to explore additional essential characteristics that could enhance their capacity to identify a broader spectrum of cyber threats. In their study, Chou and colleagues [261] harnessed AI algorithms via TensorFlow to educate the system. They equipped it with regulations and markers to differentiate between typical and anomalous network conduct. The researchers crafted a DL model on the TensorFlow platform, weaving together numerous tiers of nonlinear attributes. They then instructed the system to grasp the standard behavior through a forward propagation technique, employing the NSL-KDD dataset as a foundation. The outcomes demonstrated considerable promise, indicating remarkable test accuracy rates of 97.65 % for identifying probing assaults and 98.99 % for recognizing DDos attacks. Subsequent endeavors should concentrate on enhancing the training attributes within TensorFlow. The previous models encountered difficulties in forecasting incursions where attackers attempt to gain unauthorized access by masquerading as regular users and intrusions where attackers aim to exploit network susceptibilities for unauthorized access. This limitation stemmed from the dataset's monotonous nature, ultimately resulting in excessive learning. Dutta and colleagues [262] introduced a collective deep architecture designed for the identification and categorization of irregularities, operating effectively on both network and host tiers. The utilized datasets encompassed IoT-23 [263], LITNET2020 [183], and NetML-2020 [264], while the DL methodologies employed consisted of DNN and LSTM. As a method for feature engineering, the researchers employed a deep and sparsely constructed autoencoder (DSAE), while they adopted a stacked ensemble learning strategy for the classification task. Subsequent assessments were carried out across three diverse datasets, and the researchers deduced that the proposed approach surpassed the performance of solitary classifiers and *meta*-classifiers like RF and SVM. In their forthcoming research, the scholars propose conducting experiments on more intricate datasets, along with the utilization of advanced computational techniques to enhance processing velocity. Sun and colleagues [265] developed a model for categorizing network traffic, employing DL techniques with a specific emphasis on web and data

flows. The dataset employed to train their proposed model was curated by the researchers themselves, gathered by intercepting network traffic through different platforms. Probabilistic Neural Network (PNN) was utilized in the analysis giving an accuracy of 88.18 %, with a partition of 7:3 ratio for training and testing.

Some scholars have directed their attention towards exploring the impact of diverse network data portrayals on the efficacy of intelligent models. Millar and associates [266] conceived and compared three distinct techniques for representing network data when classifying malevolent traffic for DL applications: the content of packet payloads, graphical flow representations, and statistical flow summaries. They demonstrated the ability to forecast malicious categories with a mere 50 bytes of payload data from network packets. Given that DL thrives on extensive, extensive datasets, the researchers opted for the UNSW-NB15 dataset as the foundation for their experimentation. Among the approaches, the payload-based method demonstrated superior performance. However, all methods exhibited shortcomings in accurately identifying DDOS attacks. In light of the diverse characteristics exhibited by various malevolent assaults, no universal solution exists for recognizing all types of attacks. Consequently, in future endeavors, the researchers propose to explore a fusion of payload-based and statistical inputs as a means to identify malicious network traffic. Yang and colleagues [267] embarked on a quest to forge a system capable of identifying malevolent activities within an encrypted network, employing DL as their arsenal. The model they set forth hinged on the foundation of a Residual Neural Network (ResNet), which had the innate prowess to autonomously discern distinctive attributes while adeptly isolating the surrounding contextual information enveloping encrypted network traffic. Moreover, their endeavors were underpinned by the deployment of the CTU-13 dataset for model training. During the preliminary data preparation phase, the dataset underwent a series of transformations. Furthermore, they harnessed DQN reinforcement learning to engineer adversarial samples of encrypted traffic. The outcome was nothing short of remarkable, with the model achieving a striking accuracy rate of 99.94 %. Xu and co-authors [268] introduced a novel framework harnessing ML to bolster the detection of malware in conjunction with hardware assistance. Their approach centered on scrutinizing memory access patterns while categorizing them, all with the aim of monitoring and classification. To attain a virtual address trace, they advocated for in-processor monitoring. This trace was then dissected into distinct segments, termed “epochs,” with each epoch’s memory access patterns distilled into a set of distinctive characteristics. Subsequently, these features were funneled into ML classifiers, specifically the RF and LR. Upon comprehensive analysis, it was ascertained that the RF classifier

emerged as the most proficient choice, proving itself adept at identifying both kernel rootkits and memory corruption attacks. Remarkably, its performance was astounding in kernel rootkits detection, boasting an above 95 % average detection rate. De Lucia and collaborators [269] introduced an innovative approach for detecting malevolent network traffic concealed within encrypted data. This method employed a dual-pronged strategy, integrating SVM and CNN. In the pursuit of their experiments, the research team capitalized on a publicly available dataset [270], which encompassed a mixture of malignant and benign network data packets. The process of data preprocessing entailed the extraction of pertinent attributes from the captured packets [271]. A 70–30 % split was enacted. Both approaches exhibited notable accomplishments, boasting elevated F-scores and accuracy rates. Notably, SVM outperformed CNN, securing a superior performance in this regard. Researchers also ventured into identifying malicious activity within Virtual Private Network (VPN) data transmissions. Miller et al. [272] unveiled a computational framework, aimed at transcending the current limitations in detecting threats in VPN traffic. The devised framework harnessed a NN trained through a Multilayer Perceptron (MLP) resulting in an accuracy of more than 93 %. Wang et al. [273] showcased the efficacy of the NB model as a tool for classifying nefarious websites. They devised a self-learning system engineered to categorize websites based on their intrinsic characteristics. Utilizing the NB classifier, this system segregated websites into two discrete categories: suspicious or benign. The bedrock of their analysis rested upon the ISCX2016 dataset [172], a trove boasting a repository of over 100,000 URLs, each endowed with a rich array of 50 distinct features. Following an intricate process of feature extraction and dataset refinement, the researchers achieved an impressive accuracy rate, reaching up to 90 %. This outcome underscored the NB classifier's proficiency in the domain of website classification. Ongun and colleagues [274] turned their attention to the CTU-13 dataset, embarking on the creation of composite models geared towards the identification of suspicious network activity. To craft these models, they harnessed LR, RF, and GB. Their initial approach centered on a connection-level representation, wherein features were extracted directly from the unprocessed connection logs. Their work yielded a remarkable AUC score of 99 %. Al-Qatf et al. [275] and Niyaz and colleagues [276] embarked on a quest to engineer a versatile and streamlined NIDS. Their similar approach demonstrated an exceptional level of accuracy of an impressive 98 % performance rate. In alignment with the concept of employing DL for the task of intrusion detection, Zhang and colleagues [277] introduced a methodology anchored in the utilization of the NSL-KDD dataset. Resulting in an F-score and recall values of 76.47 % and 79.47 %, respectively. Wu and Guo [278] fused CNN and RNN, and put the new algorithm to the test using the NSL-KDD and UNSW-NB15 datasets. In the domain of binary classification, the model exhibited stellar performance, boasting an average accuracy rate of 99.24 % on the NSL-KDD dataset and an impressive 97.40 % accuracy on the UNSW-NB15 dataset. Meanwhile, in multiclass classification, the results were equally commendable, with an average accuracy score of 99.05 % on the NSL-KDD dataset and 84.98 % on the UNSW-NB15 dataset. In their pursuit of efficient network intrusion detection, Hasan and colleagues [279] harnessed the prowess of ANN to the task of distinguishing between malicious attacks and benign network connections utilizing the DARPA 1998 intrusion detection dataset [161]. They achieved attack detection accuracy rate of 92 %. Similarly, Devikrishna et al. [280] deployed ANN for the classification between the ordinary and the anomalous data within a dataset sourced from the KDD-99 repository. Table 8 shows a summary of some of the most used algorithms in detecting cybersecurity threats in the literature.

7.2. Utilized algorithms

An array of different ML and DL algorithms were utilized to evaluate their performance on a single dataset.

Table 16
Areas of Future work.

Area	Importance	Approach
Validating Models with New Datasets	Generalization: New datasets help in evaluating the generalizability of ML models across different environments and attack types.	Cross-Domain Testing: Apply models trained on data from one set of IoT devices to another set, potentially from different manufacturers or use cases, to test the adaptability and transferability of the learned models.
	Robustness: Testing with diverse data exposes the models to a variety of attack signatures and normal behaviors, thereby enhancing their robustness.	Temporal Validation: Validate models against data collected from different time periods to ensure that they remain effective as attack methods evolve.
	Adaptability: New datasets can include recent and evolving attack vectors that were not present in the original training data, challenging models to adapt to the changing landscape of cybersecurity threats.	Anomaly Detection Enhancement: Integrate anomaly detection techniques that do not rely solely on known attack signatures but can also identify novel behaviors indicative of possible threats.
Exploring the Impact of IoT Device Type and Function	Device-Specific Models: Different IoT devices (e.g., cameras, thermostats, routers) exhibit unique operational patterns and may face different types of security threats. Understanding these can lead to more specialized and effective models.	Feature Engineering: Develop and refine features that are particularly indicative of attacks in specific types of devices. For instance, a feature effective in detecting anomalies in video feed data from cameras might be different from those relevant for data traffic from a smart thermostat.
	Performance Optimization: Knowledge of how device type and function influence detection can lead to optimizations that reduce false positives and false negatives, enhancing overall system reliability.	Device-Specific Anomalies: Focus on identifying what constitutes normal and abnormal behavior for each type of device. This involves deep dives into the operational profiles of different device classes.
	Response Time Reduction: Tailoring models to specific device characteristics can potentially decrease the time needed to detect and respond to attacks, critical for mitigating damage.	Scalable and Dynamic Models: Design scalable machine learning models that can adjust their parameters based on the device type automatically. Implement dynamic models that evolve based on continuous learning from the device's operational data.

7.2.1. ML algorithms

7.2.1.1. Logistic regression (LR). LR is a statistical method used for binary classification tasks. It's a fundamental technique in ML, particularly useful when the outcome to be predicted is dichotomous, meaning it has two possible categories. For example, it can be used to determine whether an email is spam or not spam, or if a patient has a particular disease or not. The core idea behind LR is to find a relationship between features (independent variables) and the probability of a particular outcome (dependent variable). Unlike linear regression, which predicts a continuous output, logistic regression estimates the probability of a binary outcome. The logistic function, also known as the sigmoid function, is what sets logistic regression apart. This function takes any input value from the independent variables and outputs a value between

0 and 1. This output is interpreted as the probability of the dependent variable belonging to a particular category. For example, in a medical diagnosis application, the LR model might take in various patient data points (like age, symptoms, test results) and output a probability of the patient having a certain disease. If the probability is higher than a certain threshold (often 0.5), the model might classify the patient as having the disease. LR models are easy to implement, interpret, and very efficient to train, making them a popular choice for many binary classification problems. Despite their simplicity, they can provide a strong baseline for binary classification tasks.

7.2.1.2. Decision trees (DT). DT are a popular ML method used for both classification and regression tasks. This approach involves creating a model that predicts the value of a target variable by learning decision rules inferred from the data features. The model is represented as a tree structure, visually resembling a flowchart, where each internal node denotes a test on an attribute, each branch represents the outcome of that test, and each leaf node holds a label or value. The construction of a DT starts from the root node and involves dividing the data into subsets that contain instances with similar values (homogeneous). This process of splitting is based on certain criteria like Gini impurity or information gain in classification problems, and variance reduction in regression. As a result, the tree segregates the dataset into various branches, eventually leading to decision nodes that give us the prediction. DT are intuitive and easy to interpret, making them highly accessible for decision-making processes. They can handle both numerical and categorical data and are capable of performing feature selection, picking the most informative features for making splits. This makes them a powerful tool for data analysis, particularly when clarity and transparency in the decision-making process are required. However, DT are also prone to overfitting, especially when they grow deep with many branches. They can become complex and tend to memorize the training data, losing their generalization capability. To mitigate this, techniques like pruning (trimming the branches of the tree) are employed. Despite their simplicity, DT can form the basis for more advanced models like RF and GB algorithms, where multiple trees are combined to produce more accurate and robust predictions. In summary, while DT are straightforward and interpretable, their tendency to overfit and sensitivity to data variations are significant considerations in their application.

7.2.1.3. Random forest (RF). RF is an ensemble learning method predominantly used for classification and regression tasks. It operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. The fundamental concept behind RF is to combine multiple DT in determining the final output rather than relying on individual DT. This method addresses some of the limitations of DT, particularly overfitting to the training set. By aggregating the results of various trees, it reduces the risk of overfitting while maintaining the predictive power of the trees. Each tree in a RF is built from a sample drawn with replacement (i.e., a bootstrap sample) from the training set. Furthermore, when splitting a node during the construction of the tree, the split that is chosen is no longer the best split among all features. Instead, the split that is picked is the best split among a random subset of the features. This strategy of random feature selection contributes to the diversity among the trees in the forest, leading to more robust overall predictions. The RF algorithm has gained popularity due to its simplicity, effectiveness, and scalability, making it applicable to a wide range of data sets and problems. It is also relatively robust to outliers and can handle both categorical and numerical data, which adds to its versatility in practical applications.

7.2.1.4. Stochastic gradient descent (SGD). SGD is an iterative method for optimizing an objective function, typically used in ML and DL for training models. It is a variant of gradient descent, a fundamental

algorithm in optimization theory, distinguished by its use of a single training example (or a small batch of examples) to perform updates on the model parameters in each iteration, rather than the entire dataset as in traditional gradient descent. The term “stochastic” refers to the method’s inherent randomness in selecting training examples to compute the gradient of the objective function. In each iteration of SGD, a sample (or a small batch of samples) is randomly selected from the training dataset, and the gradient of the objective function is computed based on these samples. The model parameters are then updated in the direction that minimizes the objective function, using a learning rate to control the size of the update. One of the primary advantages of SGD over traditional gradient descent is its computational efficiency, especially with large datasets. By using only a subset of the data at each step, SGD can make progress in optimizing the parameters without waiting to compute the gradient over the entire dataset, which can be computationally expensive. However, the stochastic nature of the algorithm can also lead to more variability in the path towards the minimum of the objective function. This can be both a strength and a weakness: it allows SGD to potentially escape local minima more effectively than standard gradient descent, but it can also make the convergence path more erratic. To mitigate this, various modifications and improvements to the basic SGD algorithm have been proposed, such as introducing momentum to smooth out variations, or adapting the learning rate over time.

7.2.1.5. K-Nearest neighbors (KNN). KNN is a non-parametric, instance-based learning algorithm used in pattern recognition for classification and regression. In both cases, the input consists of the k closest training examples in the feature space. In KNN classification, the output is a class membership. An object is classified by a majority vote of its neighbors, with the object being assigned to the class most common among its k nearest neighbors (k is a positive integer, typically small). If $k = 1$, then the object is simply assigned to the class of that single nearest neighbor. KNN is a type of lazy learning, where the function is only approximated locally and all computation is deferred until function evaluation. The algorithm itself is straightforward, a number of k and a distance metric are chosen. To classify a new point, the distances from this point to all points in the training set are computed. The k nearest points in the training set are identified. For classification, the most frequent class among these k points is returned as the prediction. One of the key advantages of KNN is its simplicity and effectiveness, especially in cases where the decision boundary is very irregular. However, it can suffer from high computational cost, as it requires storing the entire dataset and computing distances between the new point and all other points in the dataset. Also, KNN’s performance can significantly degrade with high-dimensional data (a phenomenon often referred to as the “curse of dimensionality”).

7.2.1.6. Naïve Bayes (NB). NB is a family of simple, yet effective, probabilistic classifiers based on applying Bayes’ theorem with strong (naïve) independence assumptions between the features. It is particularly suited for high-dimensional datasets and is a popular model for text classification problems, such as spam filtering and sentiment analysis. The core principle behind NB is Bayes’ theorem, which describes the probability of an event, based on prior knowledge of conditions that might be related to the event. In the context of classification, Bayes’ theorem is used to estimate the probability that a given sample belongs to a particular class, based on the observed attributes of the sample. The ‘naïve’ aspect of the algorithm comes from its assumption that the features (or predictors) are independent of each other, given the class. This means that the presence (or absence) of a particular feature is unrelated to the presence (or absence) of any other feature, given the class. This assumption, while rarely true in real-world data, simplifies the computations involved and often leads to surprisingly good performance in many practical scenarios. NB classifiers can be trained very efficiently in

a supervised learning setting. Despite their simplicity, NB classifiers have worked quite well in many complex real-world situations. They require a small amount of training data to estimate the necessary parameters (probabilities) for classification. A Gaussian NB Assumes that the continuous values associated with each class are distributed according to a Gaussian distribution.

7.2.1.7. Linear discriminant analysis (LDA). LDA is a statistical method used in ML for both classification and dimensionality reduction. Rooted in the principles of statistics and linear algebra, LDA is particularly known for its role in pattern classification and as a technique for finding a linear combination of features that best separates two or more classes of objects or events. The fundamental idea of LDA is to model the difference between the classes of data. It does this by determining a linear combination of features that separates two or more classes of objects or events as much as possible. This is achieved by maximizing the ratio of the between-class variance to the within-class variance in any particular data set, thereby guaranteeing maximal separability. In more technical terms, LDA seeks to project the data onto a lower-dimensional space with good class-separability in order to avoid overfitting (“curse of dimensionality”) and also reduce computational costs. The LDA projection maximizes the distance between the means of the different classes while minimizing the scatter (i.e., variance) within each class. LDA is widely used in various fields, including ML, pattern recognition, and image processing, due to its simplicity and effectiveness. However, it assumes that the data is normally distributed, the classes have identical covariance matrices, and the means of the distributions are different. While these assumptions do not always hold true in real-world data, LDA can still perform quite well in practice.

7.2.1.8. Quadratic discriminant analysis (QDA). QDA is a statistical technique used in ML for classification. It is closely related to LDA, but while LDA assumes that the different classes share the same covariance matrix, QDA allows each class to have its own covariance matrix. This makes QDA a more flexible approach than LDA, as it can model a wider range of datasets, particularly those where the assumption of identical covariance matrices for each class does not hold. In QDA, the decision boundaries between the classes are quadratic, which allows for more complex relationships between the features and the class labels. The basic principle of QDA is to model the probability distribution of each class as a Gaussian distribution, with its own mean and covariance matrix. When a new data point is to be classified, the algorithm calculates the probability of that point belonging to each class and assigns it to the class with the highest probability. QDA can be more powerful than LDA in cases where the class distributions are significantly non-linear or the classes have different covariance structures. However, it also requires estimating more parameters (the covariance matrices for each class), which can be a drawback when dealing with small training datasets, as it can lead to overfitting. Therefore, the choice between LDA and QDA often depends on the size of the dataset and the underlying distribution of the features.

7.2.1.9. Extreme gradient boosting (XGB). XGB is an advanced implementation of GB algorithms, widely used in the field of machine learning. Developed by Tianqi Chen, it has gained popularity due to its efficiency, flexibility, and effectiveness in a wide range of predictive modeling tasks. GB is an ensemble technique that builds models in a stage-wise fashion. It involves training DT one at a time, where each new tree helps to correct errors made by the previously trained tree. XGB specifically applies this technique with several enhancements and optimizations, making it a powerful tool for ML competitions and practical applications. XGB has been used to win numerous ML competitions and is widely used in industry for predictive modeling tasks. Its success is attributed to its speed and performance, as well as its versatility in handling various types of data and tasks.

7.2.1.10. Adaptive boost (AdaBoost). AdaBoost is an ensemble learning algorithm that is used primarily for classification tasks. Introduced by Yoav Freund and Robert Schapire in 1995, it has since become one of the most popular boosting techniques due to its simplicity, effectiveness, and the ease with which it can be combined with various types of learning algorithms. The fundamental concept of AdaBoost is to sequentially apply a classification algorithm to repeatedly modified versions of the data, thereby producing a sequence of weak classifiers. These classifiers are then combined into a weighted sum that represents the final output of the boosted classifier. The data modifications at each iteration consist of applying weights to each of the training samples. Initially, these weights are all set equally, but with each subsequent iteration, the weights of incorrectly classified instances are increased. Thus, AdaBoost focuses on the examples that are harder to classify. AdaBoost is sensitive to noisy data and outliers. In such cases, the algorithm tries to fit the noisy data and outliers, which can lead to overfitting. Despite this, AdaBoost has been found to be very effective, especially when the underlying weak learners are moderately weak (i.e., they perform slightly better than random guessing).

7.2.1.11. Categorical boosting (CatBoost). CatBoost is an open-source GB library, particularly known for its efficiency with categorical data. Developed by researchers at Yandex, it is designed to handle categorical features automatically and is well-suited for a wide range of data science problems and ML tasks. CatBoost has an innovative gradient-boosting scheme. It uses oblivious trees as base predictors, which are symmetric trees where all nodes at the same level test the same condition. This structure allows for efficient execution and model application. One of the standout features of CatBoost is its ability to naturally handle categorical data without the need for extensive pre-processing. Traditional ML models require categorical data to be converted into numerical form, often through binarization via one-hot encoding. CatBoost, however, can process categorical data internally, which simplifies the data preparation process and often leads to better performance. CatBoost has been successfully applied in various domains such as finance, healthcare, marketing, and others, where it has been used to develop models that require handling a mix of categorical and numerical features.

7.2.1.12. Light gradient boosting Machine (LightGBM). LightGBM is a GB framework that uses tree-based learning algorithms. It is designed to be distributed and efficient with the following advantages: faster training speed and higher efficiency, lower memory usage, better accuracy, support of parallel and distributed learning, and capable of handling large-scale data. Developed by Microsoft, LightGBM extends the GB algorithm by introducing two novel techniques: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). These techniques allow LightGBM to be highly efficient, particularly on large datasets, and to outperform other types of GB models in terms of speed without compromising accuracy. In real-world datasets, there are often features that are mutually exclusive, meaning they never take non-zero values simultaneously. EFB bundles these exclusive features into a single feature, thus reducing the number of features and improving efficiency. GOSS is a novel method for filtering out the data instances to find a split value during tree growth. GOSS keeps all the instances with large gradients and performs random sampling on the instances with small gradients. This approach maintains the accuracy of learning while reducing the number of data instances to be considered. LightGBM allows users to customize the optimization objective and metric for evaluation, making it adaptable to a wide range of tasks beyond standard classification and regression problems. Due to these advantages, LightGBM is widely used in data science competitions and practical applications, particularly where the speed of training and the efficiency of memory usage are critical.

7.2.1.13. Support vector Machine (SVM). SVM is a powerful and

versatile supervised ML algorithm, primarily used for classification and regression tasks. Introduced in the 1960s and later refined in the 1990s, SVM has become a staple in the ML community due to its effectiveness in handling high-dimensional data and its versatility in solving various types of problems. The fundamental idea behind SVM is to find a hyperplane that best separates the classes in the feature space. SVM seeks to maximize the margin between the hyperplane and the nearest data points from each class, known as support vectors. This maximization of the margin is key to SVM's robustness and effectiveness in classification. For linearly separable data, SVM finds a linear hyperplane that separates the classes. However, in cases where the data is not linearly separable, SVM uses a technique called the kernel trick to transform the data into a higher-dimensional space where a linear separator can be found. The advantages of SVM include its effectiveness in high-dimensional spaces, its versatility in handling both linear and non-linear data, and its robustness against overfitting, especially in high-dimensional space.

7.2.1.14. Multi-Layer Perceptron (MLP). MLP is a type of ANN used in ML for classification tasks. It is implemented in the popular ML library, scikit-learn, in Python. An MLP consists of at least three layers of nodes: an input layer, a hidden layer, and an output layer. Except for the input nodes, each node is a neuron that uses a nonlinear activation function. MLP works by simulating the way a human brain processes information. It uses a supervised learning technique called backpropagation for training. The MLP consists of multiple layers of nodes. Each layer is fully connected to the next layer in the network. Nodes in the input layer represent the input data. All other nodes map inputs to outputs by a linear combination of the inputs with the node's weights and bias and applying an activation function such as Sigmoid. The weights are usually optimized via SGD. This can help the network learn complex patterns in data. MLP is versatile and can be used for various classification tasks, but it requires careful tuning of parameters, such as the number of hidden neurons, layers, and regularization terms. It's also important to scale the input data (e.g., to have zero mean and unit variance) before training an MLP, as it is sensitive to feature scaling.

7.2.2. DL algorithms

7.2.2.1. Convolutional Neural network (CNN). CNN is a type of DL algorithm primarily used for processing data with a grid-like topology, such as images. CNNs have been highly successful in various applications, particularly in image and video recognition, image classification, medical image analysis, and Natural Language Processing (NLP) applications. The architecture of a CNN is designed to automatically and adaptively learn spatial hierarchies of features from input images. This is achieved through the use of multiple layers that process and transform the input data. The convolutional layers perform a convolution operation, applying a set of learnable filters to the input. Each filter activates certain features from the input image, such as edges, textures, or specific objects, and produces a feature map as output [311]. Towards the end of the network, fully connected layers are used where every input is connected to every output (similar to the traditional NN). These layers are typically used to classify the features learned by the convolutional layers into various categories. CNNs are particularly powerful in capturing the spatial and temporal dependencies in an image through the application of relevant filters. The architecture ensures that the relation between pixels is well maintained, allowing these networks to capture the temporal and spatial dependencies in an image.

7.2.2.2. Fully convolutional network (FCN). FCN is a type of NN architecture that is specifically designed for semantic segmentation in the field of computer vision. Semantic segmentation refers to the process of classifying each pixel in an image into a specific category, which is crucial for various applications such as autonomous driving, medical image analysis, and image editing. The key characteristic of FCNs is that

they are composed entirely of convolutional layers, without any fully connected layers. This design allows FCNs to accept input images of any size and produce correspondingly-sized output, where the output is typically a segmented image that classifies each pixel into a category. Unlike traditional CNNs used for classification, FCNs convert these fully connected layers to convolutional layers, enabling the network to output spatial maps instead of classification scores. These maps are then upsampled to produce dense pixel-wise output. FCNs are trained end-to-end and pixels-to-pixels on semantic segmentation tasks. They have been a significant advancement in the field of computer vision, enabling efficient and accurate pixel-level segmentation. The success of FCNs in image recognition tasks is one of the key reasons for the DL revolution in the field of AI. Their ability to learn feature hierarchies makes them highly versatile and efficient for a wide range of applications beyond just image processing.

7.2.2.3. Recurrent Neural network (RNN). RNN is a type of NN that is particularly well-suited for processing sequences of data, making it a fundamental tool in the field of DL for tasks involving sequential inputs such as time series analysis, NLP, speech recognition, and more. Unlike traditional NN, which assume that all inputs (and outputs) are independent of each other, RNNs are designed to recognize the sequential characteristics of data and use patterns across time to make predictions. In essence, an RNN has a looping mechanism that allows information to persist across time steps. When an RNN processes a sequence, each element in the sequence is passed through the same network of neurons, and the output of each step is dependent on the previous steps. This recurrent nature enables the network to exhibit dynamic temporal behavior and to process input sequences of varying lengths. However, standard RNNs have limitations, particularly in learning long-range dependencies within a sequence. This is due to the vanishing gradient problem, where the gradients used in the training process (through backpropagation) can become very small, effectively preventing the network from learning correlations between distant events in a sequence. This limitation led to the development of more advanced RNN architectures, such as LSTM networks and GRUs, which are designed to better capture long-term dependencies. Despite these challenges, RNNs have been successfully applied in numerous applications. In NLP, they are used for tasks like text generation, sentiment analysis, and machine translation. In the field of finance, they are applied to predict stock prices and market trends based on historical data. In healthcare, RNNs assist in analyzing patient data over time for predictive diagnostics. In summary, RNNs represent a significant advancement in the ability to process sequential data and have opened up new possibilities in various fields that require an understanding of data across time. Their development and evolution continue to be a vibrant area of research in the quest to model sequential dependencies more effectively.

7.2.2.4. Gated Recurrent unit (GRU). GRU is a type of RNN architecture that is widely used in the field of DL, particularly for tasks involving sequential data. Introduced by Cho et al. in 2014, GRUs were developed as an alternative to LSTM networks, with the goal of simplifying the model while retaining the ability to capture dependencies in sequence data effectively. The GRU is designed to solve the vanishing gradient problem that plagues standard RNNs. This problem makes it difficult for a traditional RNN to learn and remember information over long sequences, limiting its effectiveness in many practical applications. GRUs address this issue through their unique structure, which includes gating mechanisms. GRUs have two gates, the update gate and the reset gate. These gates determine how much of the past information (from previous time steps) needs to be passed along to the future. This mechanism helps the GRU to capture dependencies over various time scales and to mitigate the vanishing gradient problem. Compared to LSTMs, GRUs have a simpler structure as they combine the forget and input gates into a single update gate. They also merge the cell state and hidden state, and as a

result, have fewer parameters. This simplicity can lead to faster training times and reduced computational complexity, making GRUs a more efficient alternative in some cases. GRUs are particularly effective in applications where the LSTM's additional complexity does not provide a significant advantage. GRUs help in modeling audio sequences. They are also applied in time series analysis, where they can predict future values based on past data. GRUs represent a streamlined alternative to LSTMs, offering a balance between complexity and performance in capturing temporal dependencies in sequential data. Their design allows them to be more computationally efficient while still effectively addressing the challenges of learning from sequences, making them a popular choice in various deep learning applications.

7.2.2.5. Long Short-Term memory (LSTM). LSTM networks are a type of RNN architecture that is particularly well-suited for learning from sequences of data. They were introduced by Hochreiter and Schmidhuber in 1997 to address the limitations of traditional RNNs, particularly in learning long-range dependencies. LSTMs are designed to remember information for long periods, which makes them ideal for a variety of sequence-based tasks such as time series prediction, NLP, speech recognition, and more. The key features that distinguish LSTMs from standard RNNs are their specialized structure and the use of gating mechanisms. The core of an LSTM unit is the memory cell which can maintain its state over time, acting as a kind of memory. The LSTM has two states that are passed to the next time step: the hidden state and the cell state. The hidden state is similar to the output of a standard RNN, while the cell state is unique to LSTMs and helps in preserving long-term dependencies. LSTMs can learn and remember over long sequences of data, which is challenging for traditional RNNs due to problems like vanishing or exploding gradients. The introduction of LSTMs represented a significant advancement in the field of DL, particularly for sequence modeling tasks. Their ability to capture long-term dependencies in data has made them a cornerstone in many state-of-the-art systems in various domains.

7.2.2.6. Attention based long Short-Term memory (ALSTM). ALSTM refers to a sophisticated NN architecture that combines the strengths of LSTM networks with an attention mechanism. This combination is particularly powerful for tasks involving sequential data where the relevance of different parts of the sequence can vary significantly, such as in NLP, speech recognition, and time series analysis. The attention mechanism, introduced in the context of NN for machine translation, allows a model to focus on the most relevant parts of the input sequence as needed. It works by assigning different weights to different parts of the input, indicating how much focus or 'attention' the model should give to each part when making predictions or generating output. This mechanism is particularly useful in tasks where certain parts of the input sequence are more informative than others. When combined, ALSTMs leverage the sequential data processing capability of LSTMs while also benefiting from the selective focus provided by the attention mechanism. ALSTMs have been successfully applied in various domains, they improve the accuracy of predictions by focusing on critical time steps.

7.2.2.7. Restrictive Boltzmann Machine (RBM). RBM is a type of stochastic ANN that plays a significant role in the field of DL. Developed initially in the context of statistical mechanics, RBMs were later adapted by Geoffrey Hinton and others for use in ML as a powerful tool for feature learning and dimensionality reduction. RBMs can learn to represent complex, high-dimensional data in a lower-dimensional space, making them useful for feature extraction in tasks like image and speech recognition. RBMs have been applied in recommendation systems to predict user preferences based on observed interactions.

7.2.2.8. Deep Belief network (DBN). DBN is a type of ANN that is particularly notable in the field of DL. DBNs contributed significantly to

the resurgence of interest in NN, leading to what is often referred to as the "deep learning revolution." DBNs are generative models, meaning they can learn to probabilistically reconstruct their inputs, and they are composed of multiple layers of stochastic, latent variables. A DBN is typically composed of multiple layers of hidden units, where each layer is an RBM or a variant of it. The training of a DBN occurs in two main phases: unsupervised pre-training and supervised fine-tuning. During pre-training, each layer of the network is trained as an RBM, which allows the network to learn complex representations of the input data. After pre-training, the network undergoes a fine-tuning phase using a supervised learning technique, often backpropagation, to perform specific tasks like classification. One of the key contributions of DBNs is their ability to effectively train deep architectures. Before the advent of DBNs, training deep neural networks was challenging due to issues like the vanishing gradient problem. The layer-wise pre-training approach of DBNs helps in initializing the weights in a way that makes the subsequent fine-tuning process more effective and stable. DBNs have been applied in various domains, including image recognition, video recognition, and motion-capture data analysis. In the field of speech recognition, they have been used to learn features from raw audio data. DBNs are also employed in NLP tasks, where they can be used to learn semantic representations of text. DBNs development marked a significant milestone in the ability to train deep networks and contributed to the broader adoption and success of deep learning methods.

7.2.2.9. Probabilistic Neural network (PNN). PNN is a type of feed-forward NN that is used primarily for classification tasks and is based on the concept of Bayesian probability. Developed by Donald F. Specht in 1990, PNNs are distinguished by their ability to provide probabilistic decisions about the class membership of input data. They are particularly known for their fast-training times and their capability to produce accurate results even with a limited amount of training data. The architecture of a PNN is typically composed of four layers: an input layer, a pattern layer (also known as a hidden layer), a summation layer, and an output layer. Unlike many other NN architectures, PNNs can be trained very quickly. This is because the training essentially involves storing the training data and setting the parameters of the probability density function. Due to their probabilistic nature, PNNs are generally more robust to overfitting compared to other types of NN. PNNs can adapt immediately to new training data, making them suitable for applications where the training data is continuously updated.

7.2.2.10. Learning vector quantization (LVQ). LVQ is a type of ANN algorithm that falls under the category of competitive learning and supervised learning. Developed by Teuvo Kohonen in the late 1980s, LVQ is primarily used for classification tasks and is known for its simplicity and effectiveness in various applications, especially in pattern recognition. LVQ consists of a single layer of neurons, and each neuron represents a class. The learning process in LVQ involves adjusting the position of these neurons in the feature space to model the distribution of the training samples. During training, the algorithm iteratively adjusts the neurons (also called codebook vectors or prototypes) based on the input data and their associated class labels. In LVQ, neurons compete to respond to a given input. The neuron whose weight vector is closest to the input vector (usually measured by Euclidean distance) is declared the winner. This process is akin to the k-NN algorithm, but with the significant difference that LVQ learns the representative points (prototypes) during training. The core of the LVQ algorithm is the rule used to update the prototypes. If the winning neuron's prototype belongs to the same class as the input vector, the prototype is moved closer to the input vector. If the classes differ, the prototype is moved away. This adjustment is done using a learning rate parameter, which decreases over time. The advantages of LVQ include its simplicity, interpretability, and effectiveness in certain types of classification tasks. It is particularly useful when the data is not linearly separable. However, LVQ has

limitations in handling high-dimensional data and complex decision boundaries. Additionally, the performance of LVQ can be sensitive to the initialization of prototypes and the choice of learning rate. Table 9 list all ML, DL, and DHL algorithms that were utilized in this paper.

Setting parameters for a proposed algorithm, particularly in machine learning or any computational models, involves a crucial process that can significantly influence the effectiveness and efficiency of the algorithm. Before setting parameters, it's essential to thoroughly understand the algorithm's mechanics, such as, what each parameter does, its expected range of values, and how it interacts with other parameters. This foundational knowledge guides the parameter tuning process. The parameter settings is tailored to the specific needs and constraints of the problem at hand. This includes considerations of data characteristics, desired accuracy, computational efficiency, and any other specific outcomes desired from the algorithm. The process starts with a baseline set of parameters. These can be default values provided by the algorithm's developers, or values derived from similar problems addressed in the literature. The training dataset is then used to empirically test the algorithm with the initial parameters. This involves running the algorithm and evaluating its performance based on predefined metrics such as accuracy, precision, recall, F1 score, and so on. Finally, optimization via Grid Search or Random Search is used to finetune the algorithm. In Grid Search, it systematically varies each parameter through a predefined range of values and test the performance of the algorithm at each variant to find the optimal combination. While in Random Search, it randomly selects parameter values within a specified range and distribution, which can sometimes lead to faster convergence on an optimal solution than grid search. Other optimization techniques can be used as well [312,313,314,315,316].

7.3. Utilized dataset

The N_BaIoT dataset is a notable resource in the field of cybersecurity, particularly for research in IoT security and the detection of botnet activities. This dataset was specifically created to aid in the development and evaluation of NIDS that can identify malicious activities in IoT environments. The N_BaIoT dataset was developed to address the growing concerns around IoT security. As IoT devices become increasingly common, they are often targeted by botnets and other cyber threats. This dataset provides a basis for researchers and security professionals to develop and test systems that can detect such threats. The dataset was generated by simulating real-world IoT network environments under both normal operating conditions and under various botnet attack scenarios. It includes network traffic data from both benign and malicious sources, offering a diverse range of patterns for analysis. The malicious traffic in the dataset includes several types of botnet attacks. These might include different methods used by botnets to compromise IoT devices. The N_BaIoT dataset typically includes a wide range of features extracted from network traffic, such as packet size, timing information, protocol usage, and other network behavior indicators that are crucial for identifying potential security threats. The primary use of this dataset is in training and evaluating ML and DL models for NIDS. Researchers use it to develop algorithms that can automatically distinguish between normal and malicious network traffic in IoT environments. One of the key challenges in IoT security that the N_BaIoT dataset helps address is the detection of subtle and sophisticated botnet activities that might not be easily detectable with traditional network monitoring tools. The dataset is crucial for advancing the state-of-the-art in IoT security, helping develop more effective detection systems against a growing range of cyber threats targeting IoT devices. N_BaIoT is extensively used for applying machine learning and AI techniques in cybersecurity, providing a rich source of data for developing predictive models and anomaly detection systems. It serves as a benchmark for evaluating the performance of various intrusion detection algorithms, facilitating comparisons and improvements in the field.

The N-BaIoT dataset consisted of multiple sub-datasets that

represented the traffic flow data during attacks or normal traffic for nine commercial IoT devices infected by botnets. These devices are DANMINI video Door Phone Doorbell Hands-free Wireless Intercom, Ecobee Smart Thermostats, Ennoi Monitor video Doorbell Intercom System, Philips Baby Monitor, Provision PT737E and PT838 Security Cameras, Samsung SNH-1011 N IP Wireless Smart CAM, and SimpleHome XCS71002 and XCS71003 Security Cameras. In our analysis, only two (Samsung SNH-1011 N IP Wireless Smart CAM and Ennoi Monitor video Doorbell Intercom System) of the nine devices were excluded from the analysis due to their target feature being heavily unbalanced. For the remaining devices' datasets, no upsampling has been done. The devices were attacked by two notorious botnets Mirai and Gafgyt. The dataset contained 115 input features and 1 output feature that has three outcome categories, the Mirai and Gafgyt botnets (Attack category), and the Bening category which refers to normal network traffic flow where no attacks were observed. An 80/20 split has been used to form the training and test data sets considering the large scale of the data set.

The N_BaIoT dataset is generally made available to the research community for use in academic and security research. Accessibility details, including terms of use and how to obtain the dataset, are typically found in the corresponding research publications or the hosting institution's website. The N_BaIoT dataset represents a significant resource for cybersecurity research in the IoT domain, enabling the development of advanced security solutions to protect against increasingly sophisticated and diverse cyber threats targeting IoT devices. It underscores the importance of data-driven approaches in enhancing IoT security and provides a valuable benchmark for the cybersecurity research community.

In summary, the N_BaIoT dataset is a well-known resource in the cybersecurity field, particularly for research involving the detection of network-based attacks on IoT devices. The N_BaIoT dataset was created to facilitate the development and testing of ML models specifically designed to detect botnet attacks on IoT devices. It is used to train models to recognize patterns of malicious network traffic indicative of such attacks. The dataset contains real traffic data from various IoT devices that were subjected to a range of known botnet attacks. This includes data both from normal operations and from periods where the devices were compromised. The dataset includes multiple types of botnet attacks, making it a comprehensive resource for testing how well detection models perform across different attack vectors. The data typically include network traffic features such as packet sizes, timing, and protocol types. These features are critical for training models that perform anomaly detection by analyzing traffic patterns. Researchers utilize the N_BaIoT dataset to evaluate the effectiveness of various machine learning techniques, including supervised and unsupervised learning methods. It serves as a benchmark for comparing the performance of different algorithms in accurately identifying IoT-specific cybersecurity threats. While the N_BaIoT dataset is a valuable resource, it also poses challenges such as potential biases in the data, the risk of overfitting models to the specific attacks included, and the need to continually update the dataset with new attack types as adversaries evolve. More information about the origins of our dataset can be found in the paper that discussed the creation and collection of this dataset [317].

7.3.1. Dataset preprocessing

Feature scaling is a critical preprocessing step in ML and data analysis, particularly necessary when dealing with datasets that contain features with differing scales. This process involves adjusting the range of different features in a dataset to a common scale, typically without distorting the differences in the ranges of values or losing information. Feature scaling is essential because most ML algorithms, especially those involving distance calculations such as KNN and GB algorithms, are sensitive to the scale of input features. The primary goal of feature scaling is to ensure that no single feature dominates the model due to its larger magnitude. When features are on different scales, there's a risk

that higher weightage is given to features with larger scales, making the model biased towards those features, potentially leading to inaccurate predictions. The Standard Scaler, often known in statistics as Z-score normalization, is a feature scaling technique widely used in data preprocessing for ML. The main idea behind the Standard Scaler is to transform each feature in your dataset so that it has a mean of 0 and a standard deviation of 1, bringing it to a standard normal distribution. This is achieved by subtracting the mean of each feature and then dividing by the standard deviation. If a feature's variance is orders of magnitude more than others, it might dominate the objective function and make the estimator unable to learn from other features correctly. By standardizing features, we ensure that each feature contributes approximately proportionately to the final distance. This not only improves the performance of these algorithms but also helps in speeding up the convergence of gradient descent optimization methods, as it ensures that all the features are on a similar scale, leading to a more balanced and well-behaved training process [318].

One-hot encoding is a widely used technique in data preprocessing to handle categorical variables. In many ML models, especially those based on mathematical equations like regression models, categorical data in text form cannot be directly used. One-hot encoding addresses this by converting categorical variables into a form that could be provided to these algorithms to improve prediction and classification accuracy. The essence of one-hot encoding lies in its method of representing categorical variables as binary vectors. This process begins by identifying the unique categories in a feature column. For each unique category, a new binary column (feature) is created. In these new columns, the presence of a category in the original data is marked with a '1', while its absence is indicated with a '0'. This approach effectively transforms a single column of categorical data into multiple columns of binary values, each representing the presence or absence of a category. One significant advantage of one-hot encoding is that it removes any hierarchical order that might be incorrectly interpreted by the model. However, one-hot encoding can significantly increase the dataset's dimensionality. This can lead to increased computational complexity and memory usage, which might be problematic with large datasets.

8. Results and Discussions

All ML models were trained with and without PCA except for the NB model. PCA reduces the dimensionality of a dataset consisting of many correlated variables while retaining the variation present in the dataset, up to the maximum extent. PCA was used with Maximum Likelihood Estimator (MLE) with n component to guess the dimension reduction [319]. Also, Cross-Validation (CV) has been used on all of the ML models. We choose a value of $k = 5$ which is very common in the field of ML [320,321]. GridSearch CV or RandomizedSearch CV for hyper-parameters tuning for each ML model. The difference between both approaches is in GridSearch we define the combinations and do training of the model whereas in RandomizedSearchCV the model selects the combinations randomly. Fig. 9 below shows the confusion matrix values for the DANMINI video Door Phone Doorbell Hands-free Wireless Intercom for DT, RF, SVM, SGD, MLR, and KNN models.

Table 10 shows the average precision, recall, and accuracy values for the DANMINI Video Door Phone Doorbell Hands-free Wireless Intercom for all variations of the proposed ML models. Note that all ML models were applied with CV. And Fig. 10 and Fig. 11 show a summary of ML model accuracies when detecting Mirai and Gafgyt attacks in all the seven sensors respectively.

It can be seen from Fig. 10 that all ML models had good accuracies (above 80 %) when detecting Mirai botnet attacks except for the Multinomial NB model which performed relatively weak compared to other models. NB showed the lowest Mirai botnet detection accuracy of 40 % for the Doorbell device and showed the highest Mirai botnet detection accuracy of 87.5 % for the Simple Home 1002 Security Camera. This suggests that device type or function specifics might affect the

performance of a model. This can also be seen for LDA and QDA models' performance as well in detecting Mirai botnet attacks for the Doorbell device and the Philips Baby Monitor compared to the other remaining devices. Furthermore, combining PCA with ML models did little to no improvement in Mirai botnet detection accuracies. Overall, most of the deployed ML models showed a great performance in detecting Mirai botnet attacks with accuracies that reached up to 100 % in some cases such as XGB, LightGBM, and DT for most devices. It can be seen from Fig. 11 that the NB model had the worst performance in all devices compared to other models when detecting Gafgyt botnet attacks. NB model is fast when performing multi-class prediction. However, the MNB model with a smoothing technique has proven to still be a weak estimator and cannot outperform other ML models. At the same time, NB gave another indicator that device type and functionality could somehow affect the performance of the ML model when detecting attacks since MNB showed its highest detection accuracy of 99.3 % when detecting Gafgyt botnet attacks in Simple Home 1003 Security Camera. NB detection values for Gafgyt ranged between 61.3 % and 93.9 % which made it the lowest-performing ML model. The performance gap between LDA and QDA models with respect to other ML models (except for NB) ranges between 1 % and 30 % depending on the device type. Furthermore, all ML models (except for NB) showed slight but noticeable differences in their accuracies in detecting Gafgyt botnet attacks in Simple Home 1003 Security Camera, unlike in other devices such as PT737E Security Camera, PT838 Security Camera, Thermostat, and Simple Home 1002 Security Camera where these difference were barely noticeable. Table 10 shows that the best ML models to use are DT and XGB because both of them achieved about 100 % in terms of accuracy, precision, and recall when detecting Mirai and Botnet attacks in the Doorbell device. The same was also true for the other devices except for Philips Baby Monitor and Simple Home 1002 Security Camera which still showed great values that were close to 99 % in some cases. Table 11 below shows a summary of the average detection accuracy for Mirai and Gafgyt botnet attacks for all ML models per device.

Most models show consistent high performance across different devices, indicating robustness in their ability to generalize the detection of botnet attacks in varied IoT environments. However, there are some noticeable drops in accuracy for specific models on certain devices, such as the sharp decline observed in the ROC – based model. The ROC-based model shows a significant variance in performance, particularly displaying a drastic drop in accuracy for devices like the Philips Baby Monitor and the SH1003 Camera. This suggests that the feature set or the anomaly detection threshold set by the ROC curve may not align well with the traffic patterns or attack signatures of these particular devices. Models like RF and GB generally maintain high accuracy. These ensemble methods are effective as they combine multiple decision trees to reduce overfitting and improve generalization over different types of data inputs, which is critical in IoT settings where device behaviors can vary widely. Simpler models like DT and NB tend to perform reasonably well but are not as robust as ensemble methods. Their variability in performance can be attributed to their susceptibility to overfitting or underfitting, depending on the complexity of the dataset from each device. Each IoT device generates different types and volumes of network traffic. Models that perform well generally have a better capability to extract and learn the most discriminative features from diverse traffic patterns. More complex models, or those with a capacity for extensive feature interaction (such as GB), are often better at capturing subtle patterns in the data, which can be crucial for detecting sophisticated botnet behaviors. Some models are more robust to noise and data variability, an essential trait for IoT environments where device behavior can change based on firmware updates, user settings, and other external factors.

Table 12 shows the average precision, recall, and accuracy values for the DANMINI video Door Phone Doorbell Hands-free Wireless Intercom for all variations of the proposed DL models. Fig. 12 and Fig. 13 show a summary of DL model accuracies when detecting Mirai and Gafgyt

attacks in all seven sensors respectively.

It is evident that the ML models outperformed the DL models in terms of average accuracy, precision, and recall in detecting botnet attacks in the Doorbell device. The same is also true for the other six remaining devices. It can be seen from Fig. 12 that the DBN model had the lowest performance in detecting Mirai botnet attacks in all devices (except for the Philips Baby Monitor) compared to other DL models. Fig. 12 also shows that DL models performed slightly lower in detecting Mirai botnet attacks in Doorbell and Philips Baby Monitor devices compared with the other devices. RNN has shown a detection accuracy of less than 90 % in detecting Mirai botnet attacks in Doorbell and Philips Baby Monitor devices compared with the other devices where it made an above 97.5 % detection accuracy. In detecting Mirai botnet attacks, LVQ had an accuracy of 80 % in both Doorbell and Philips Baby Monitor devices, an 85 % in PT737E Security Camera Device, 92.5 % in Simple Home 1002 Security Camera, 95 % in Simple Home 1003 Security Camera, and almost 98 % in the last two devices. Other slight differences in model performance across different devices can be noticed in CNN-LSTM models, while at the same time, CNN had a Mirai botnet detection accuracy of 100 % in Doorbell, Philips Baby Monitor, PT838 Security Camera, and Simple Home 1002 Security Camera. Overall, but with an exception of the poorly performed models, DL models have shown up to 99 % Mirai botnet detection accuracy in several devices. It can be seen from Fig. 13, that the DBN model had the worst performance in detecting Gafgyt botnet attacks compared to other DL models. Taking a closer look at how DBN performed in detecting Gafgyt attacks in Philips Baby Monitor where less than 1 % accuracy was achieved, it can indicate that there could be a relationship between model accuracy and botnet type being detected. LVQ model recorded a Mirai botnet detection accuracy of 60 % in Doorbell and Philips Baby Monitor devices, 70 % in PT838 Security Camera Device, and closer to 90 % in the remaining devices. Even though the performance of DBN and LVQ were poor compared to other DL models in detecting Mirai and Gafgyt botnet attacks, they showed a very noticeable difference for accuracies per botnet type and across different devices. Overall and except for the DBN model for some devices, DL models have performed better in detecting Mirai than Gafgyt botnet attacks. Table 13 below shows a summary of the average detection accuracy for Mirai and Gafgyt botnet attacks for all DL models per device. Table 14 shows the average detection accuracy for all variations of ML and DL models in all IoT devices.

Models like CNN, LSTM, and their variants generally maintain high accuracy across different devices. This indicates their strong feature extraction and temporal analysis capabilities, which are crucial for identifying patterns associated with botnet attacks. The RNN-based models show significant dips in performance on certain devices like the PT737E Camera and the Thermostat. This might suggest that these models, while generally effective at temporal sequence learning, might be less robust to variations in attack signatures specific to certain device types or to the noise present in their data streams. RNN based models are highly sensitive to the input data's sequence and structure. Their performance can fluctuate based on how well the sequence and timing of data points align with the characteristic patterns of botnet attacks. More complex models, such as those involving LSTM, exhibit variability in performance. This could be due to overfitting or underfitting depending on the dataset's complexity and the model's sensitivity to the sequence length and feature dynamics of the IoT device data. Devices with more straightforward, less variable data might be easier for simpler models to handle, while those with complex, highly variable data benefit from the sophisticated architectures of CNNs and LSTMs. The representativeness of the training data for each device's normal and attack behavior plays a crucial role. If the training data does not adequately capture the diversity of normal behavior and potential attack vectors, models may not perform well during validation.

Based on Table 14, all the DHL models have outperformed most of the ML and DL models. DL and ML models have different strengths and weaknesses, and combining them in a DHL model can be a two-edged

sword. However, in some cases, it can capitalize on their complementary strengths. The performance of DHL models is highly dependent on the specific problem, dataset, and model architecture. In some cases, DHL models may outperform both standalone DL and ML models, while in others, they may perform somewhere in between. The key is to carefully design and evaluate the DHL model based on the specific requirements of the task at hand. Fig. 14 shows DHL models' accuracies when detecting botnet attacks.

Most models show high performance across different IoT devices, with values close to 1. This indicates strong model generalization across different types of IoT environments for detecting botnet attacks. There are notable drops in performance for specific models on certain devices. For instance, models like FCN_LSTM and CNN_LSTM show significant dips in value for some devices. The SH1003 Camera shows a considerable drop in performance for some models, whereas other models maintain high values. For the SH1002 Camera, similar drops are seen, indicating these models may struggle with certain data patterns from these specific cameras. Both PT838 Camera and PT737E Camera show high overall performance, but there are occasional drops for models such as CNN_LSTM, hinting at potential overfitting or underfitting issues with these models for certain data sequences. The architecture of each model, including the number of layers, types of layers (e.g., convolutional, recurrent), and activation functions, plays a crucial role in its ability to learn and generalize from the data. Models like CNN and LSTM are generally good at capturing spatial and temporal features, respectively. However, specific combinations (e.g., CNN_LSTM) might struggle with particular data characteristics. The type and volume of data generated by each IoT device can significantly impact model performance. Devices like cameras and baby monitors generate large amounts of data with possibly different noise and pattern characteristics, affecting how well models can learn from and detect anomalies. The ability of models to extract relevant features from the data is critical. Models that fail to capture important features or are sensitive to irrelevant noise may show lower performance.

8.1. Limitations

The detection of security threats in IoT domain is grappling with numerous challenges that must be targeted to ensure the continuous improvement of cybersecurity. A significant obstacle in this context is the diverse nature of IoT devices. This heterogeneity makes it difficult to develop a one-size-fits-all security solution, complicating efforts to enhance overall security in the IoT landscape. Table 15 summarizes a list of the limitations and some of the proposed solutions.

The models designed for IoT botnet detection may encounter challenges when applied to other fields such as healthcare, manufacturing, and urban infrastructure. For instance, in healthcare, the privacy concerns and stringent regulatory requirements could limit the applicability of models trained on less sensitive IoT environments. These models might not adequately address the confidentiality and integrity of sensitive patient data, nor comply with health-specific regulations. In manufacturing, the variation in machine behavior and operational conditions can influence model performance. The models developed for detecting IoT botnets might not be sensitive enough to detect anomalies that signify equipment malfunctions or quality control issues, which are often subtle and require high specificity. Additionally, in urban infrastructure management, the models may struggle with the scale and complexity of data from smart city applications. These environments are characterized by a high volume of diverse data sources, such as traffic patterns, public transportation systems, and utility usage. The models might not be sufficiently robust or scalable to effectively process and analyze this varied data without significant retraining and adaptation. Each of these sectors presents unique challenges that require tailored approaches.

8.2. Final thoughts

The influence of IoT in manufacturing is set to continuous expansion, as its advantages are likely to surpass any challenges. IoT's capacity to enhance efficiency and provide valuable insights presents a strong argument for its broad integration, revolutionizing manufacturing akin to the initial industrial revolution. However, this immense potential also brings considerable risks. Cybersecurity contributes significantly by protecting data, which in turn strengthens the company's reputation and fosters customer trust and loyalty. Additionally, it ensures stable and predictable operations, which are essential characteristics of effective manufacturing processes. The focus should be on simplicity and automation for optimizing limited security resources, much like how automation has transformed traditional manufacturing processes. Realtime intelligent automation in cybersecurity alleviates the traditional time-consuming monitoring tasks and enhances the NIDS's efficiency and quality. Consequently, this leads to a significant reduction in the average response time to security incidents, aligning well with manufacturers ultimate goals of waste reduction and efficient operations. Fig. 15 shows how cyber resilience can enhance operations continuity.

9. Conclusion and future work

IoT's role in manufacturing is poised for continued growth, with its potential benefits outweighing the drawbacks. The efficiencies and insights provided by IoT make a compelling case for its widespread adoption, fundamentally transforming manufacturing much like the original industrial revolution did. This paper carried out a successful implementation of over 25 advanced ML and DL algorithms, including DHL models, to detect botnet attacks on seven IoT devices. The primary objective was to develop robust and accurate data analytics-based framework for attack classification and detection in IoT devices for identifying security threats; and allow for a comprehensive performance benchmark for all these models utilizing the same dataset to address a gap that was found in our literature review. The results show the successful implementation of various DL and ML models to effectively identify, classify, and detect different types of botnet attacks compromising seven different IoT devices. Using the detection accuracy of each botnet attack as a performance metric to evaluate the effectiveness of each model across the seven devices, the results have shown that ML algorithms have outperformed DL algorithms. It also indicated that some approaches may work better in detecting a certain botnet over the other and the type and function of the devices being attacked could affect the performance of the model. Future work might focus on applying the same models to another emerging dataset that includes Mirai and Gafgyt attacks to verify if certain models perform better in detecting certain botnet attacks. At the same time, more research needs to be done to evaluate if the type and function of the IIoT device could affect the performance of the model and the time it takes to detect an attack. In addition, all the DHL models have outperformed most of the ML and DL models. Overall, further analysis has to be done to evaluate the effectiveness of the models when we have a multiclass target feature with more than three classes like in our case. One can also suggest the use of autoencoders with NN to see if it can enhance the accuracy of DL models compared to those of the DL models. Future research should focus on validating these models with new datasets and exploring how the type and function of IoT devices influence detection performance and response time. Table 16 provides an elaboration on this.

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CRediT authorship contribution statement

Mohammad Shahin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mazdak Maghanaki:** Writing – review & editing, Visualization, Methodology, Investigation. **Ali Hosseinzadeh:** Writing – review & editing, Methodology, Investigation, Data curation. **F. Frank Chen:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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